Detector Acceptance Study

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April 13, 2026

Overview

- Files Used & Event Selection
- Invariant Mass Studies
- x_F Studies
- p_T Studies (Yields & Acceptances)
- Track Kinematics (dp_x, dp_y)
- Binned Slice Studies (x_F and p_T bins)

Files Used & Kinematic Bins

Files Used:

- LH2 and LD2 Drell-Yan Monte Carlo (M027_S001)
- **Thrown:** `mc_drellyan_*_4pi_pTxFweight_v2.root`
- **Accepted:** `mc_drellyan_*_clean_occ_pTxFweight_v2.root`

Acceptance Calculation:

$$\text{Acceptance} = \frac{\sum w_{\text{accepted}}}{\sum w_{\text{thrown}}} \quad \text{where } w = \text{ReWeight}$$

Kinematic Bin Boundaries:

- **Mass [GeV]:** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- x_F**:** [0.0, 0.05, 0.10, ..., 0.80]
- p_T **[GeV/c]:** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]

Event Selection Criteria (Chuck's Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8$ GeV
- $-0.1 < x_F < 0.95$
- $0.05 < x_T \leq 0.58$
- $38 < dp_z < 116$ GeV
- $dp_x^2 + dp_y^2 < 5.0$ GeV^2

Target & Vertex:

- $-280 < dz < -5$ cm
- $|dx| < 0.25$ cm
- $|dy - 1.6| < 0.22$ cm

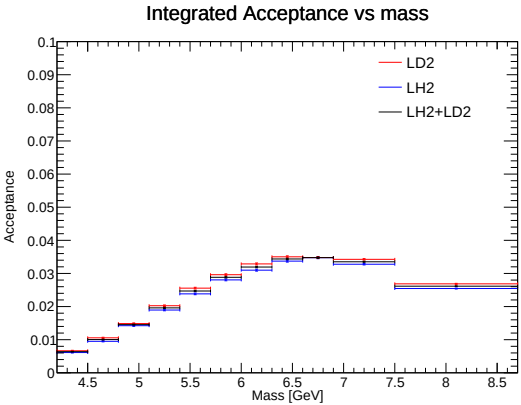
Track Quality:

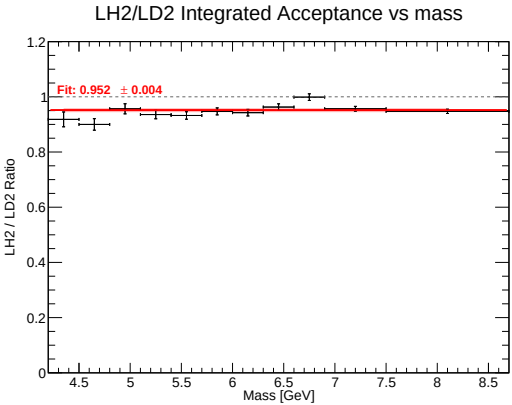
- $\chi^2_{\text{dimuon}} < 18$
- Track separation < 270 cm
- $p_{z,\text{st1}} \in [9, 75]$ GeV
- $\text{nHits}_1 > 13, \text{nHits}_2 > 13$

Generator-Level Fiducial Cuts (Thrown):

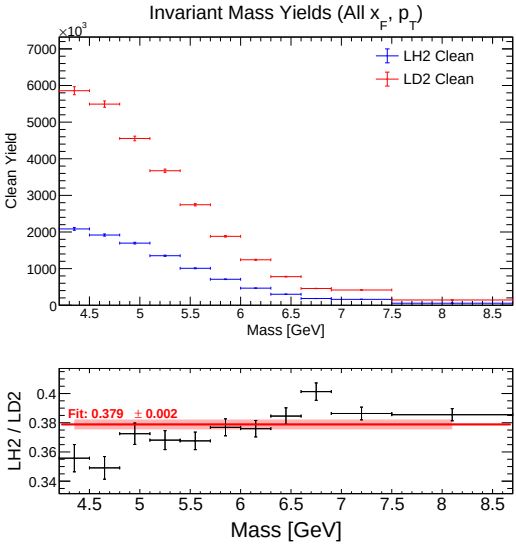
- $4.2 < \text{Mass} < 8.8$ GeV
- $-0.1 < x_F < 0.95$
- $0.0 < p_T \leq 3.0$ GeV/c

Overlay & Ratio: Mass (Integrated)

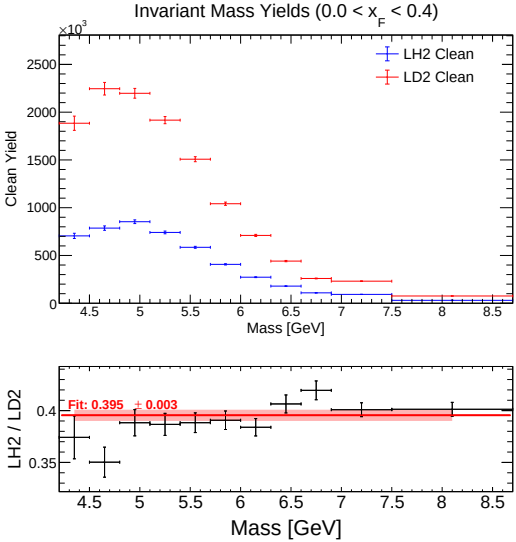




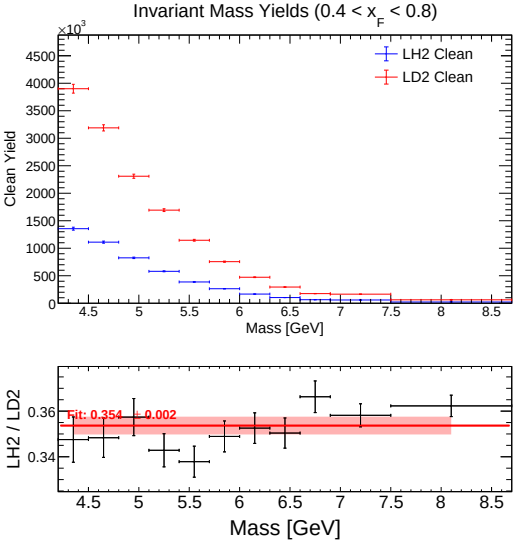
Invariant Mass Yields & Ratio (All x_F, p_T)



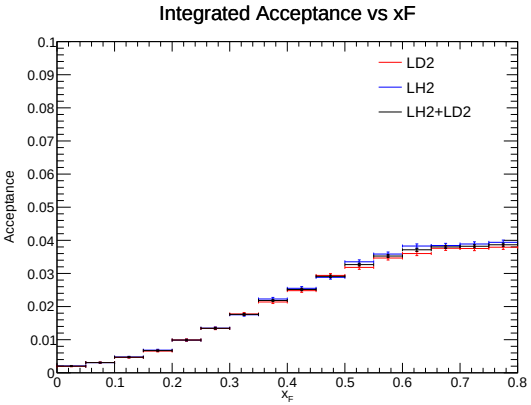
Invariant Mass Yields & Ratio ($0.0 < x_F \leq 0.4$)

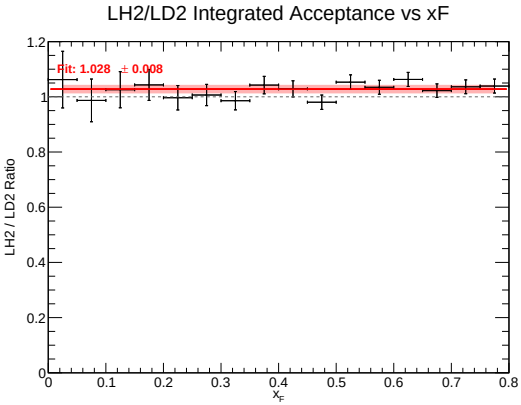


Invariant Mass Yields & Ratio ($0.4 < x_F \leq 0.8$)

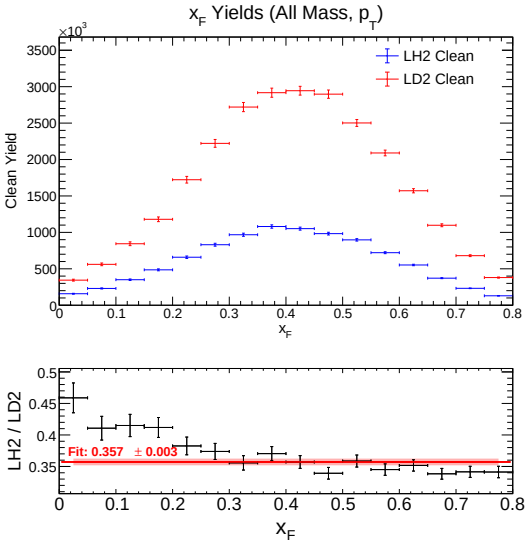


Overlay & Ratio: x_F (Integrated)

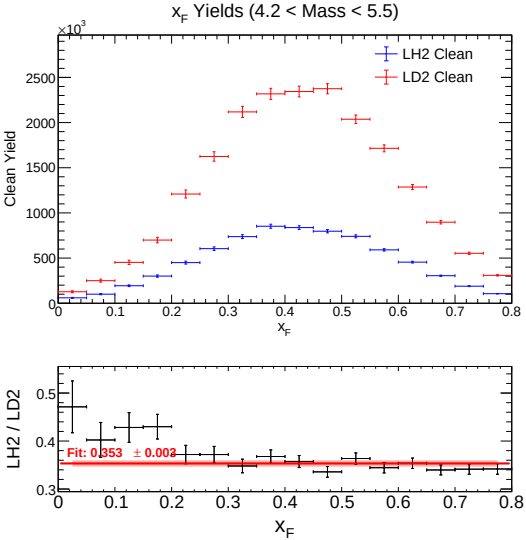




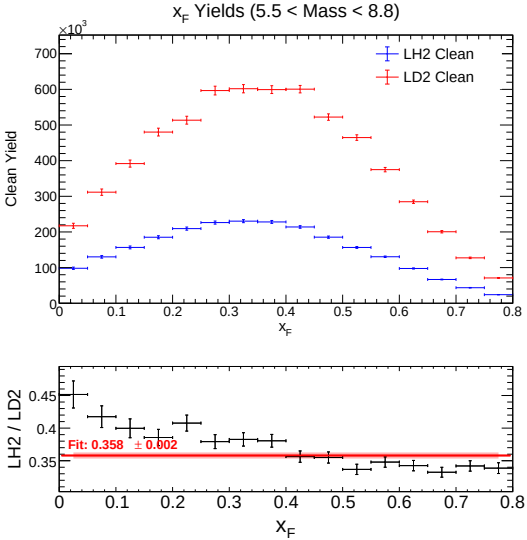
x_F Yields & Ratio (All Mass, p_T)



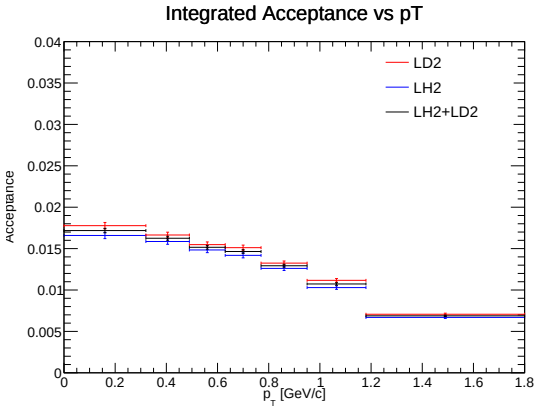
x_F Yields & Ratio ($4.2 < \text{Mass} \leq 5.5$)

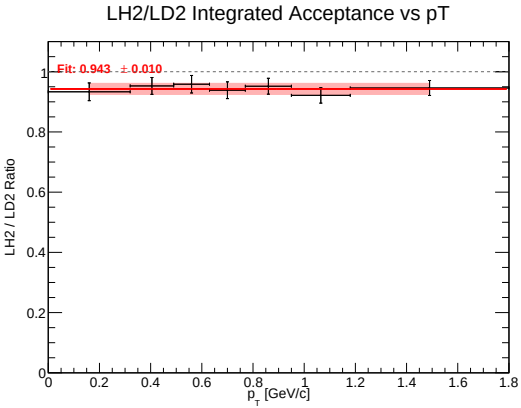


x_F Yields & Ratio ($5.5 < \text{Mass} \leq 8.8$)

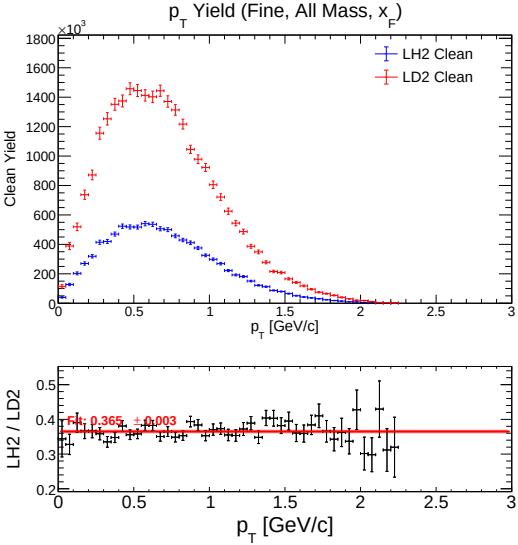


Overlay & Ratio: p_T (Integrated)

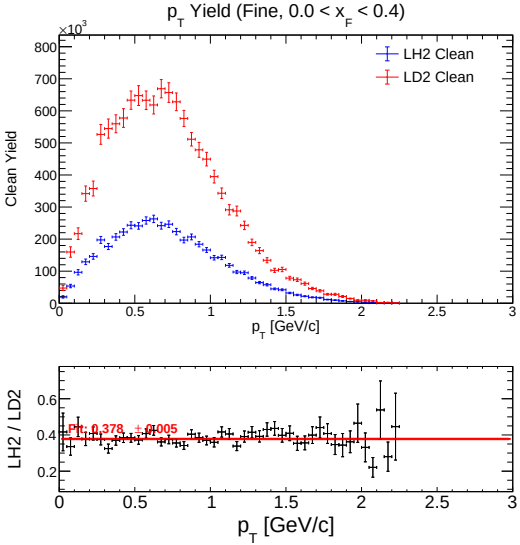




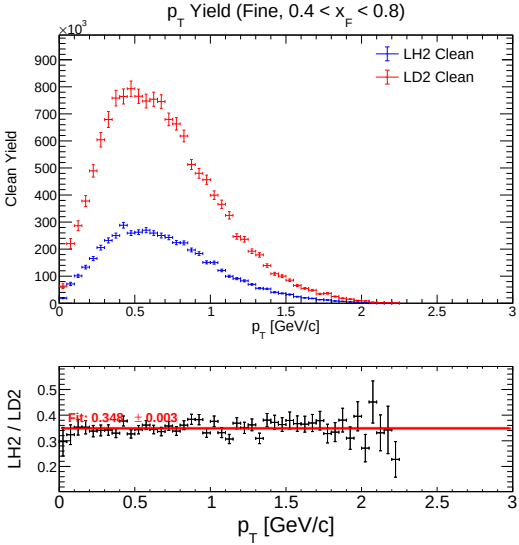
p_T Yields & Ratio (Fine Bins, All Mass, x_F)



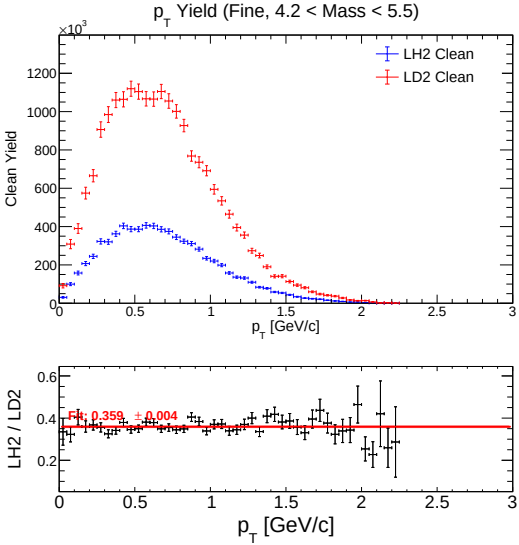
p_T Yields & Ratio (Fine Bins, $0.0 < x_F < 0.4$)



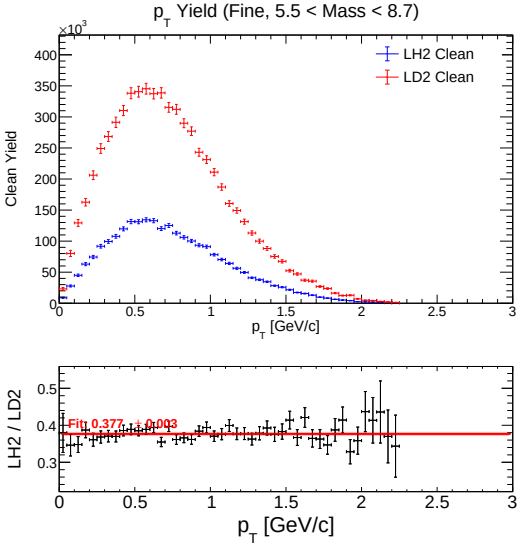
p_T Yields & Ratio (Fine Bins, $0.4 < x_F < 0.8$)



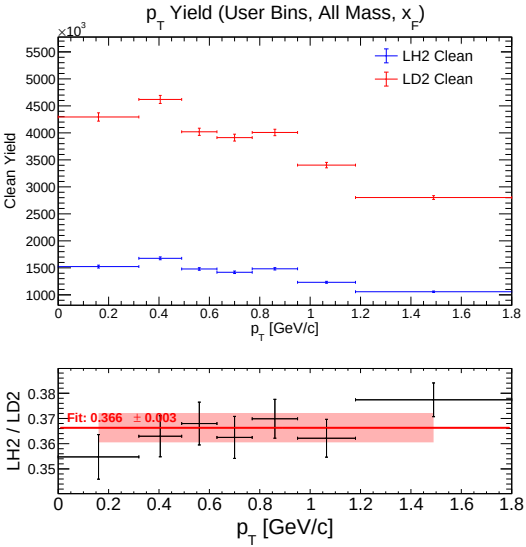
p_T Yields & Ratio (Fine Bins, 4.2 < Mass < 5.5)



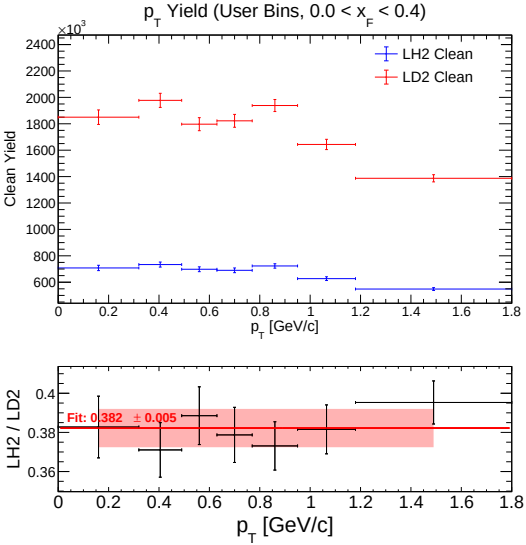
p_T Yields & Ratio (Fine Bins, 5.5 < Mass < 8.7)



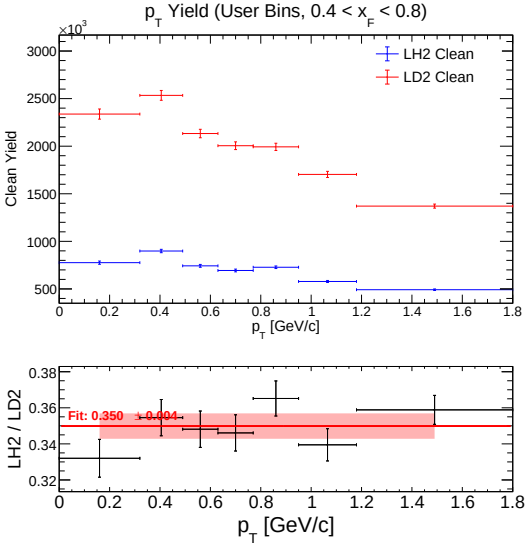
p_T Yields & Ratio (User Bins, All Mass, x_F)



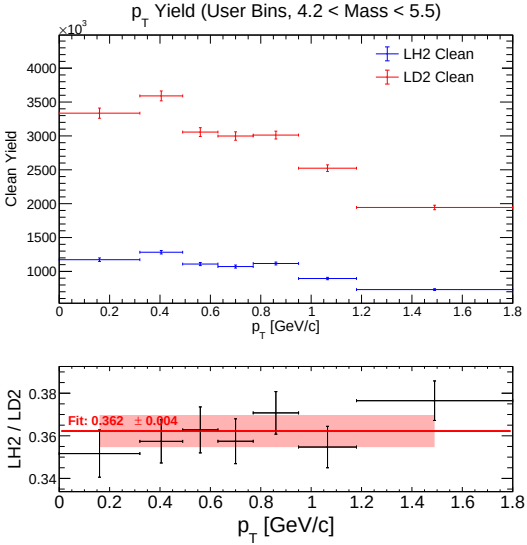
p_T Yields & Ratio (User Bins, $0.0 < x_F < 0.4$)



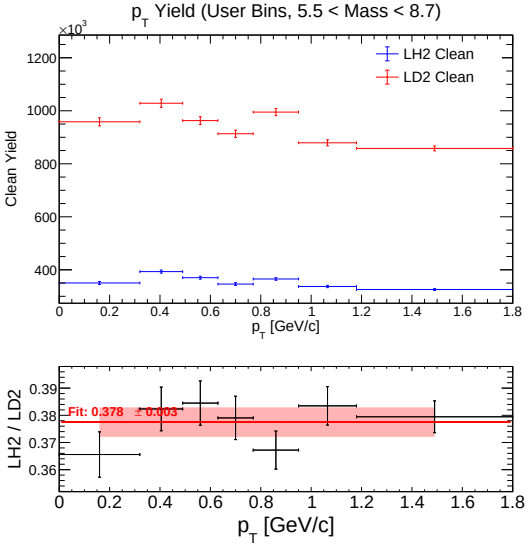
p_T Yields & Ratio (User Bins, $0.4 < x_F < 0.8$)



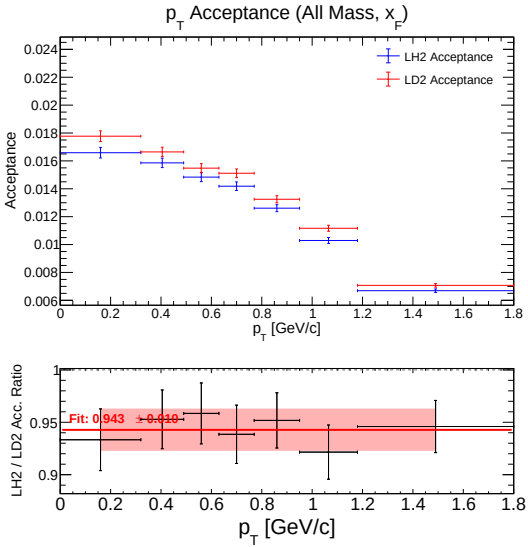
p_T Yields & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)



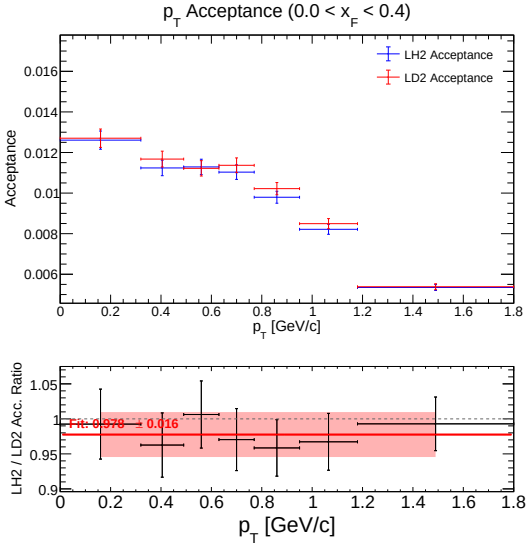
p_T Yields & Ratio (User Bins, $5.5 < \text{Mass} < 8.7$)



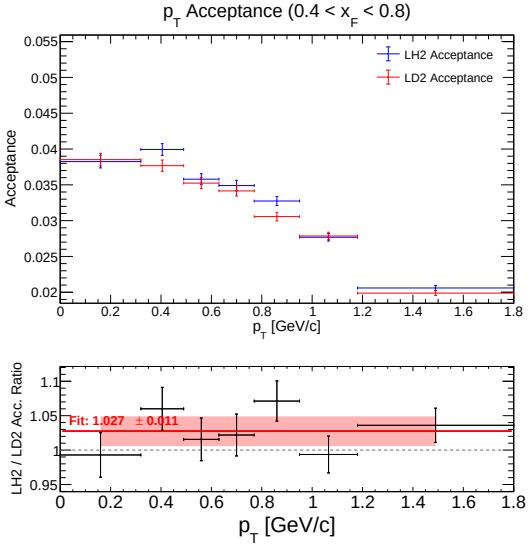
p_T Acceptance & Ratio (User Bins, All Mass, x_F)



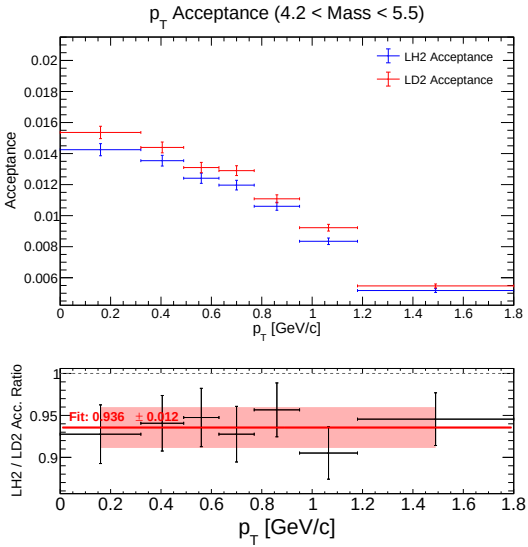
p_T Acceptance & Ratio (User Bins, $0.0 < x_F < 0.4$)



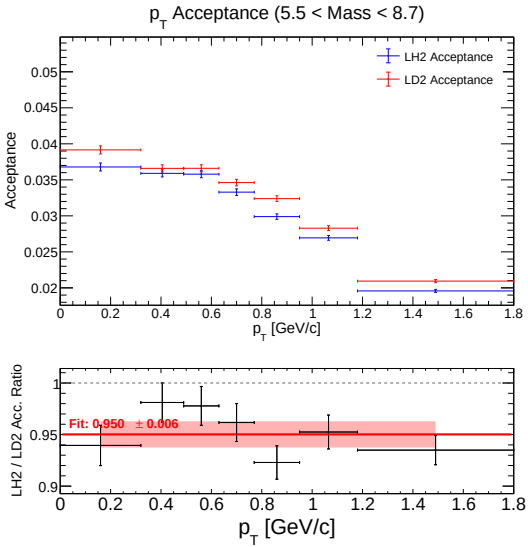
p_T Acceptance & Ratio (User Bins, $0.4 < x_F < 0.8$)



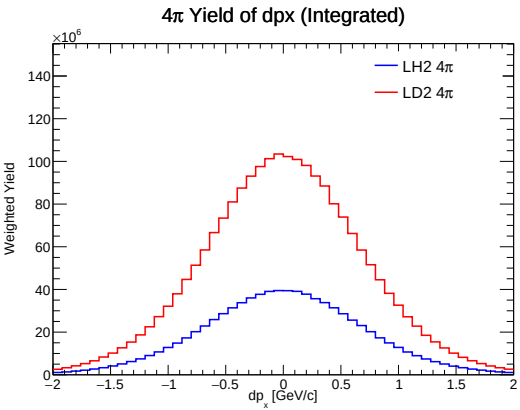
p_T Acceptance & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)

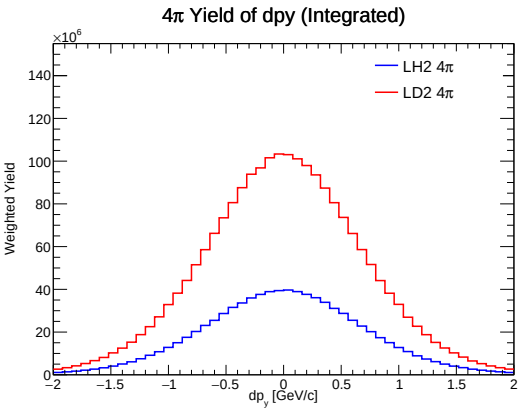


p_T Acceptance & Ratio (User Bins, $5.5 < \text{Mass} < 8.7$)

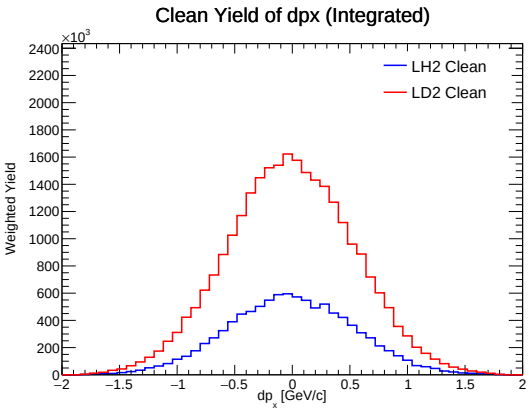


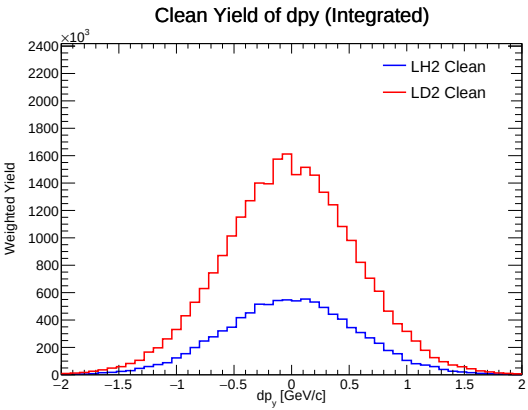
Kinematic Yields (4π Thrown): Integrated



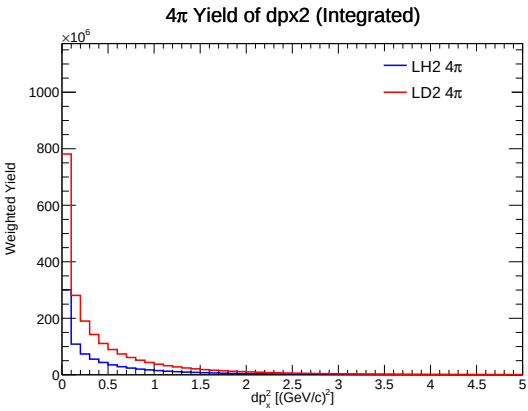


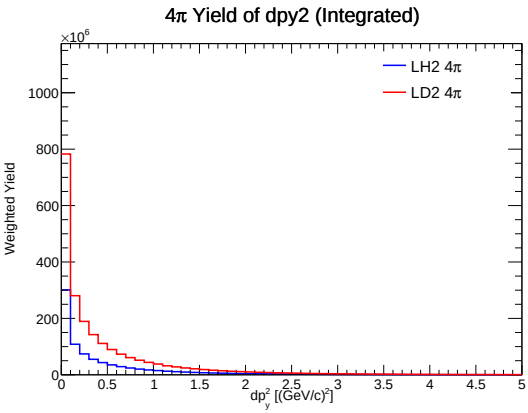
Kinematic Yields (Clean Accepted): Integrated



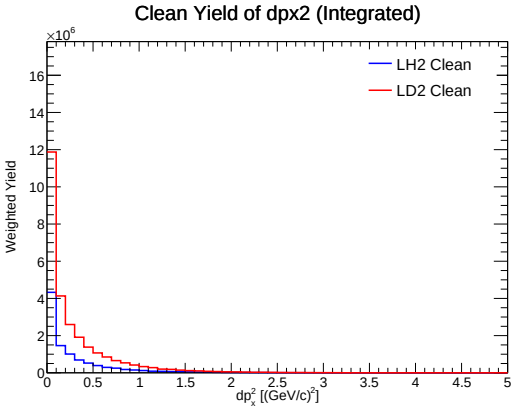


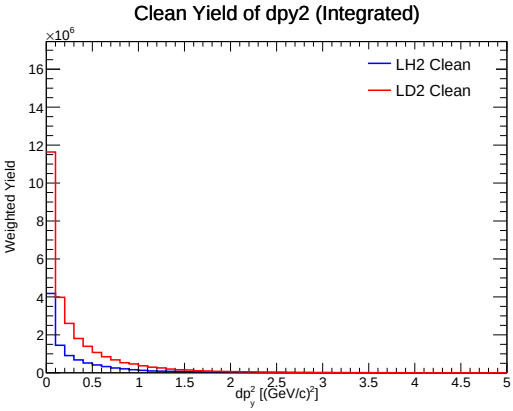
Kinematic Yields Squared (4π Thrown): Integrated



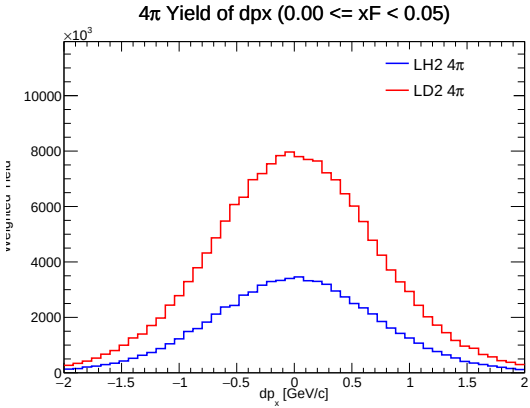


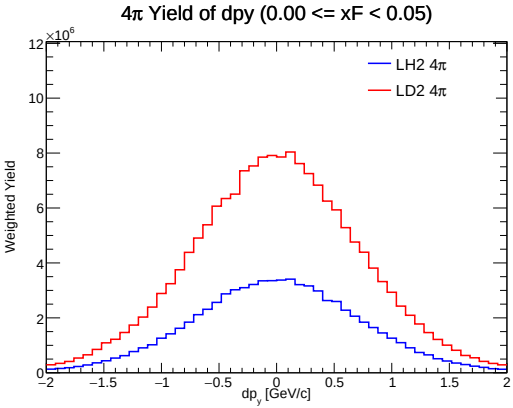
Kinematic Yields Squared (Clean Accepted): Integrated



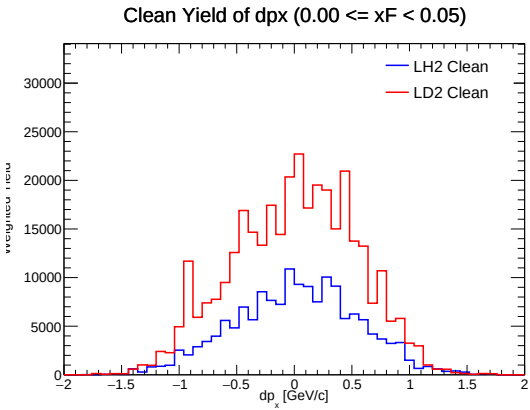


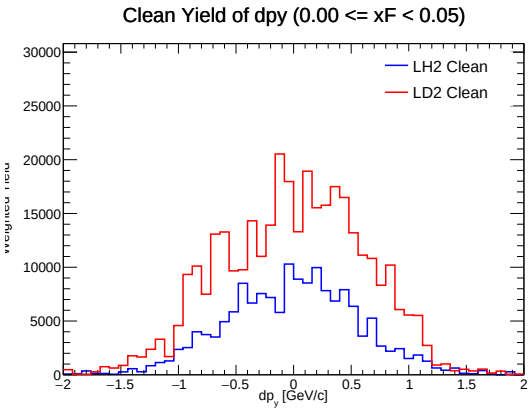
Kinematic Yields (4π Thrown): x_F Bin 0



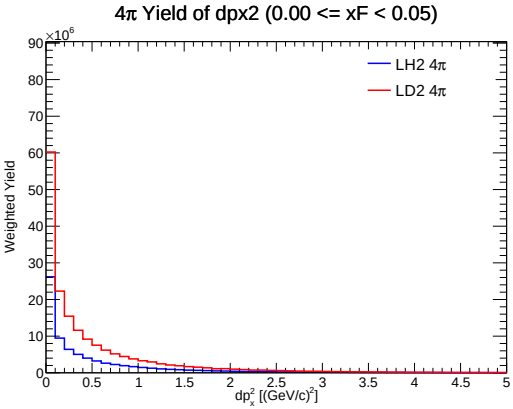


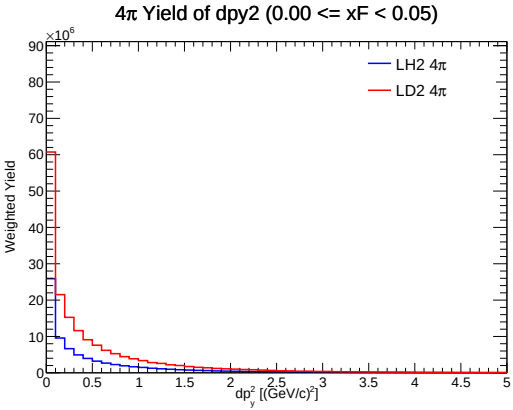
Kinematic Yields (Clean Accepted): x_F Bin 0



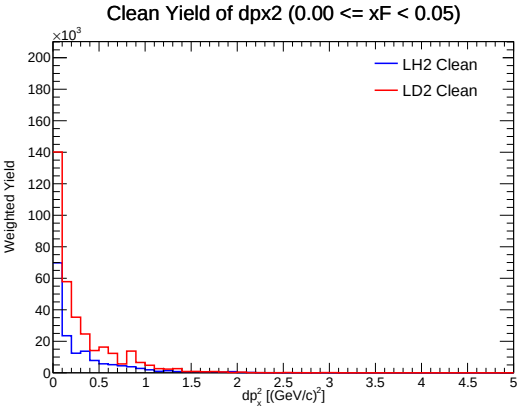


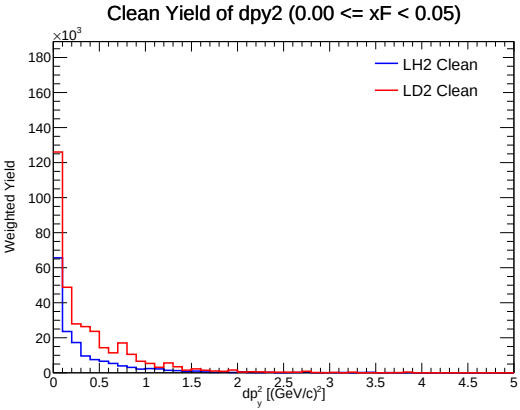
Kinematic Yields Squared (4π Thrown): x_F Bin 0



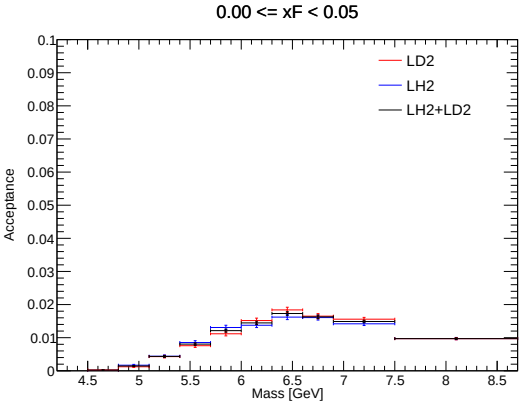


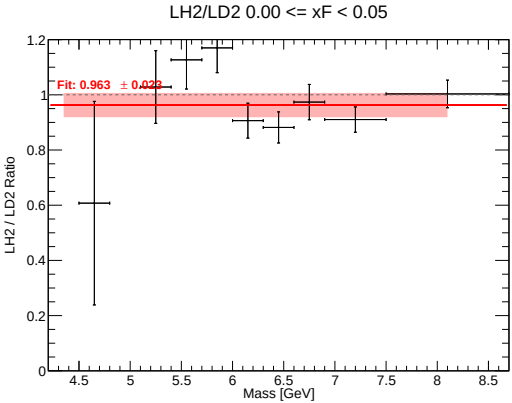
Kinematic Yields Squared (Clean Accepted): x_F Bin 0



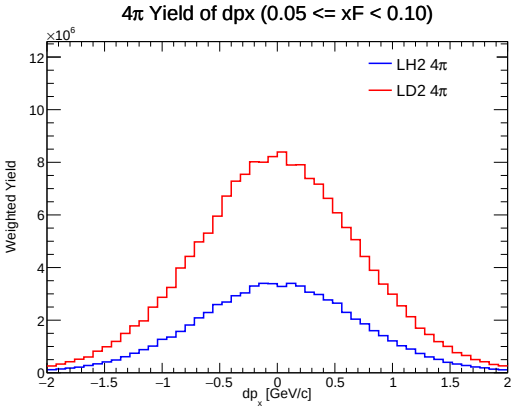


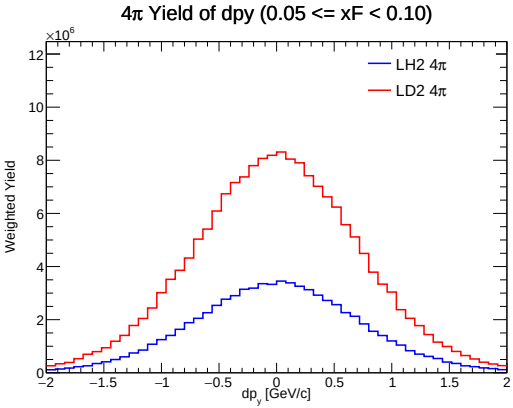
Overlay & Ratio: x_F Bin 0



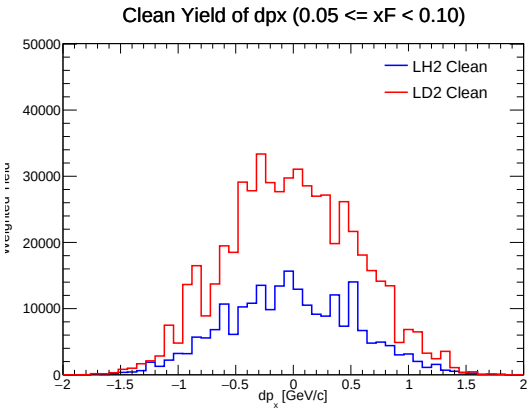


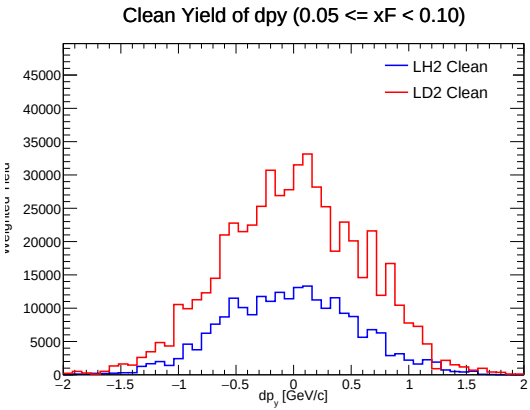
Kinematic Yields (4π Thrown): x_F Bin 1



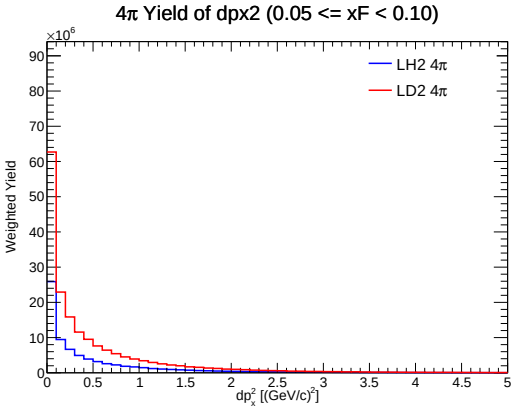


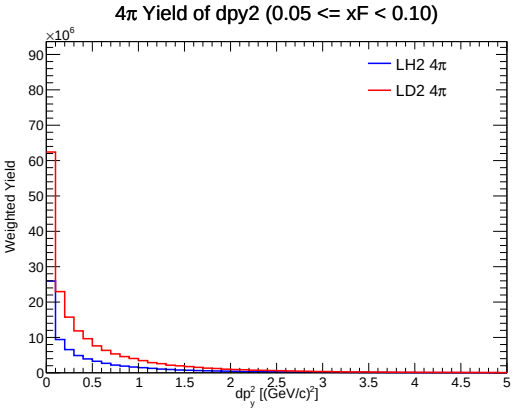
Kinematic Yields (Clean Accepted): x_F Bin 1



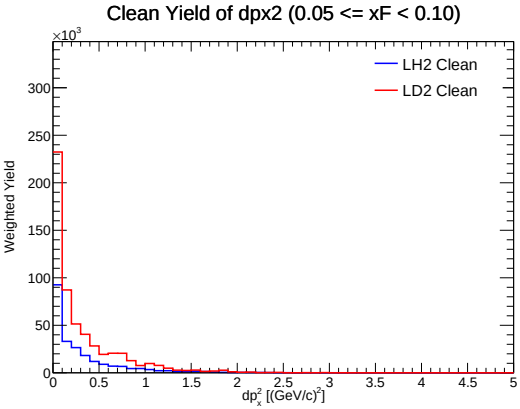


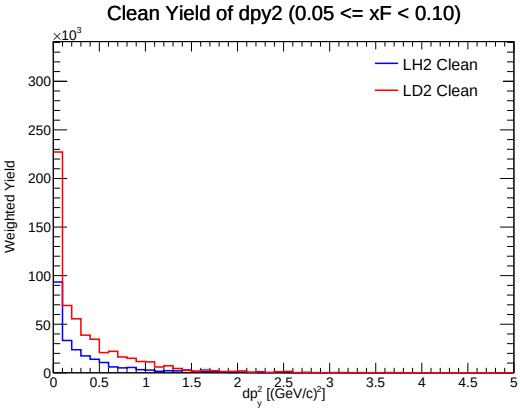
Kinematic Yields Squared (4π Thrown): x_F Bin 1



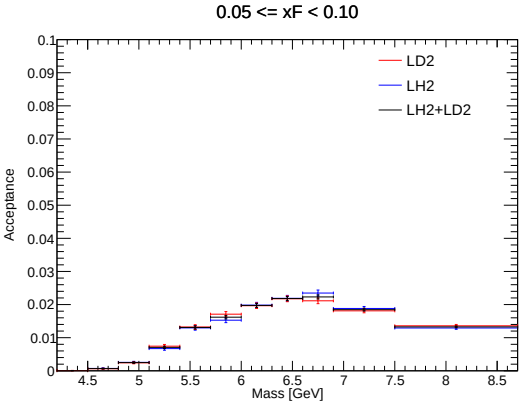


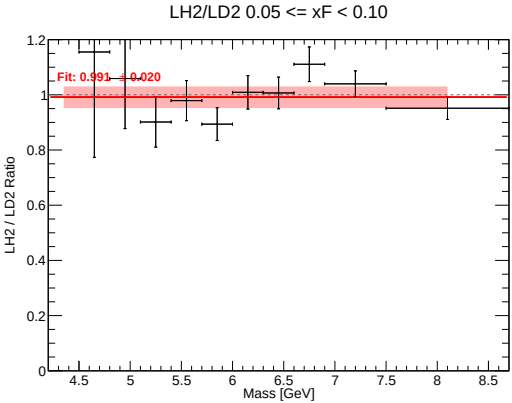
Kinematic Yields Squared (Clean Accepted): x_F Bin 1



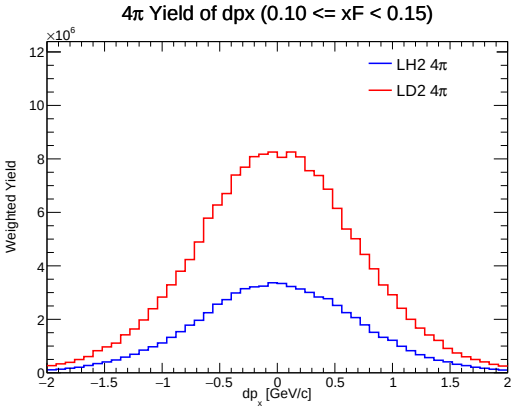


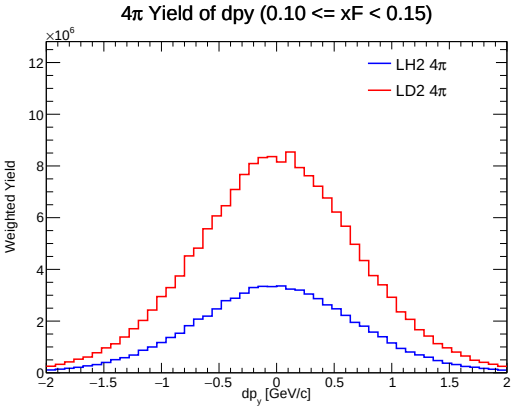
Overlay & Ratio: x_F Bin 1



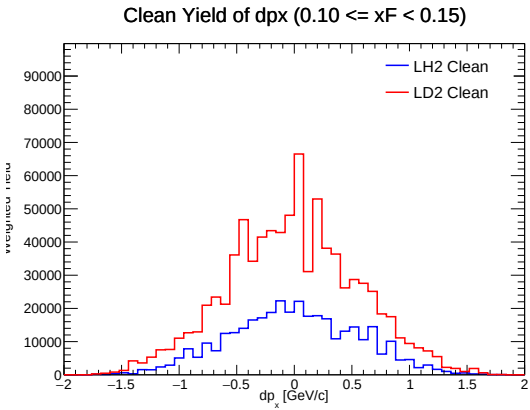


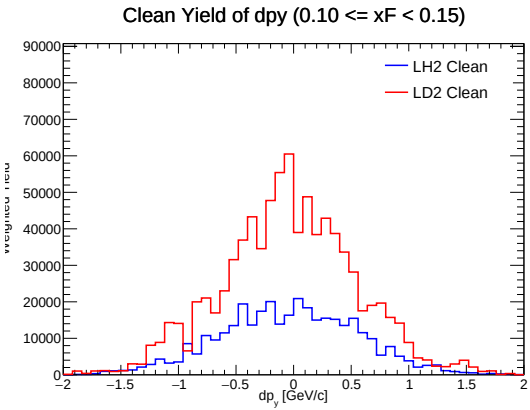
Kinematic Yields (4π Thrown): x_F Bin 2



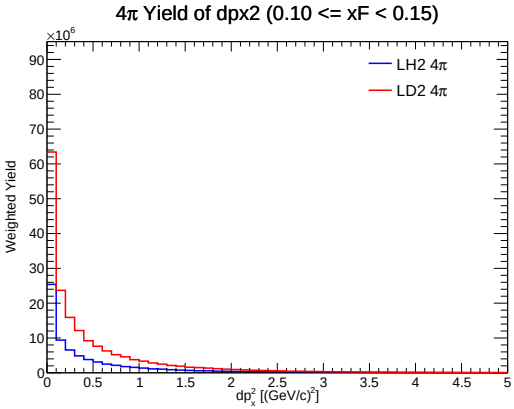


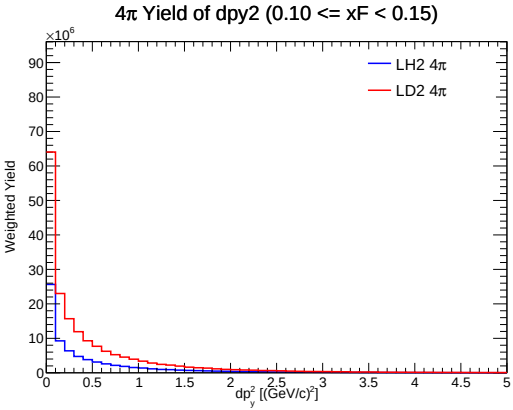
Kinematic Yields (Clean Accepted): x_F Bin 2



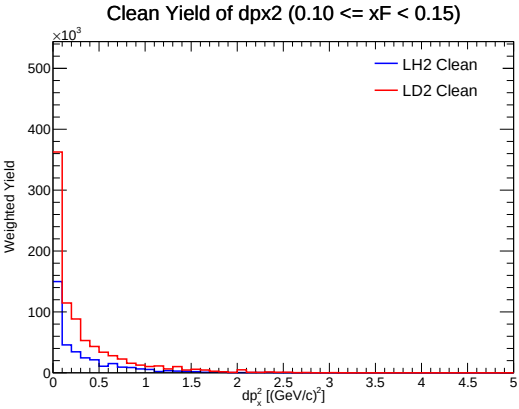


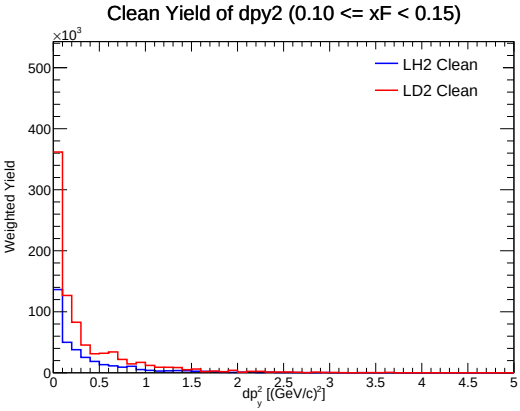
Kinematic Yields Squared (4π Thrown): x_F Bin 2



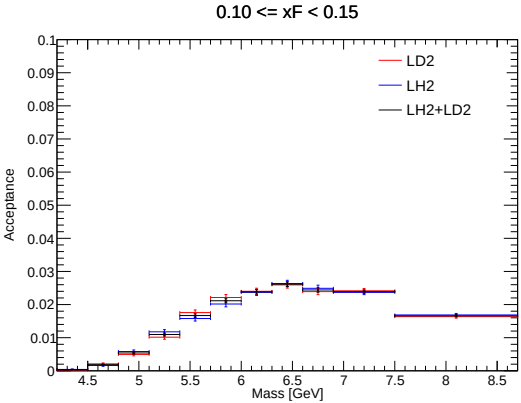


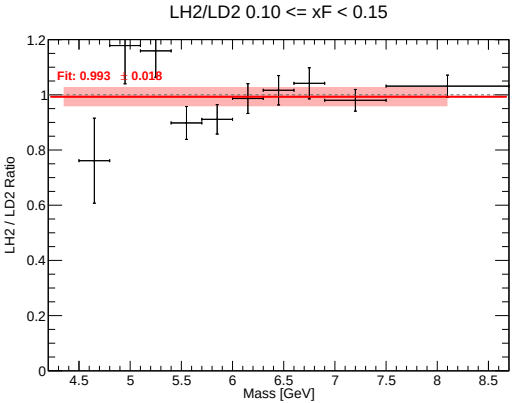
Kinematic Yields Squared (Clean Accepted): x_F Bin 2



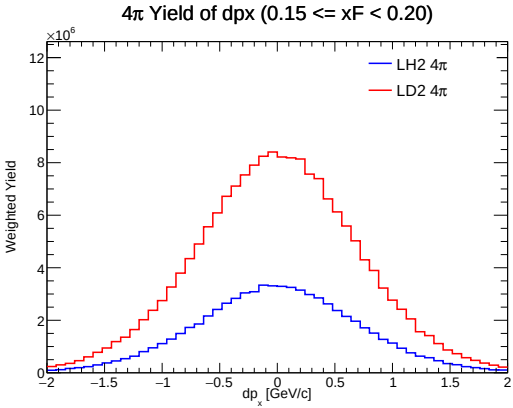


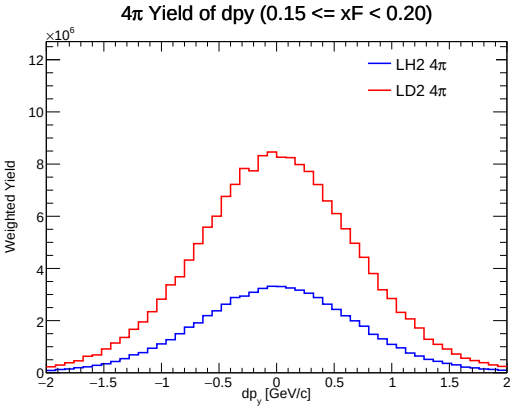
Overlay & Ratio: x_F Bin 2



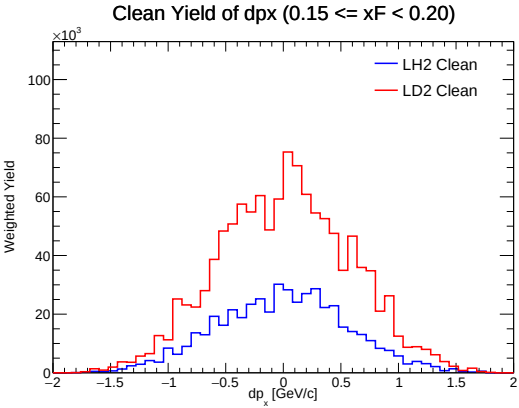


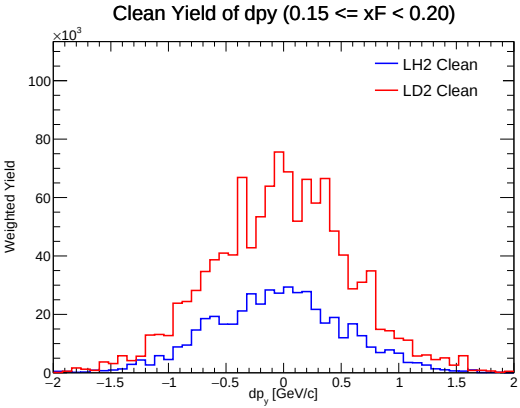
Kinematic Yields (4π Thrown): x_F Bin 3



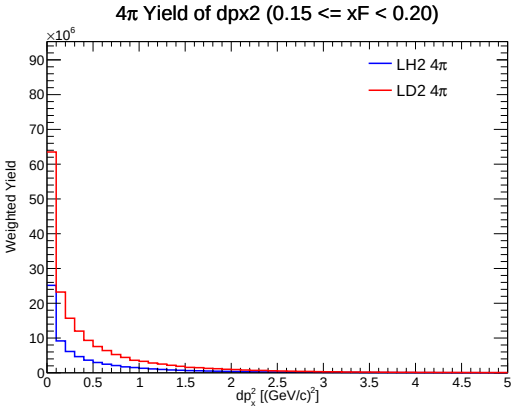


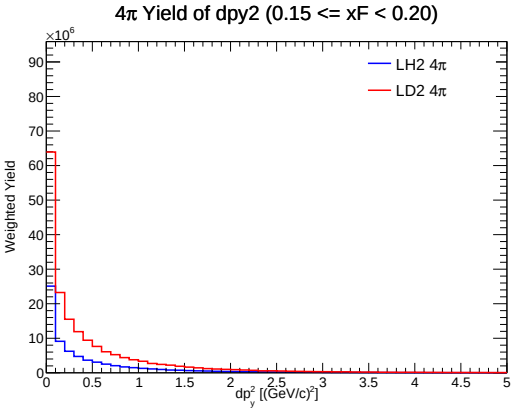
Kinematic Yields (Clean Accepted): x_F Bin 3



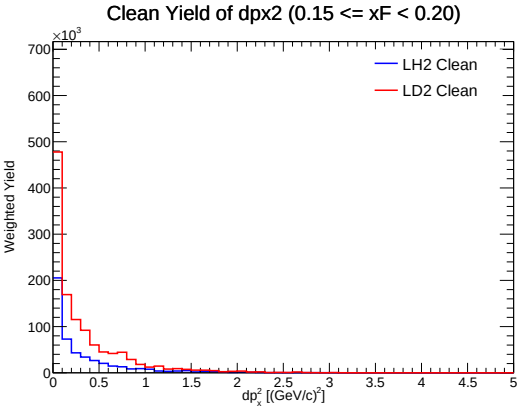


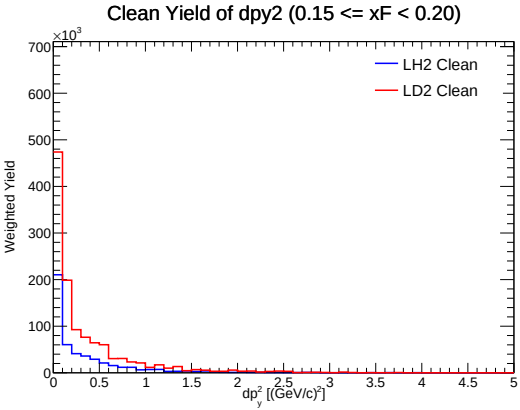
Kinematic Yields Squared (4π Thrown): x_F Bin 3



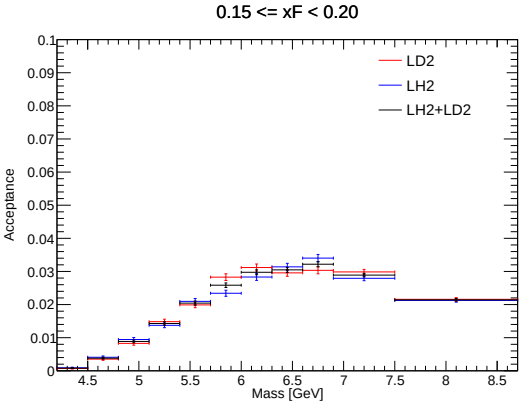


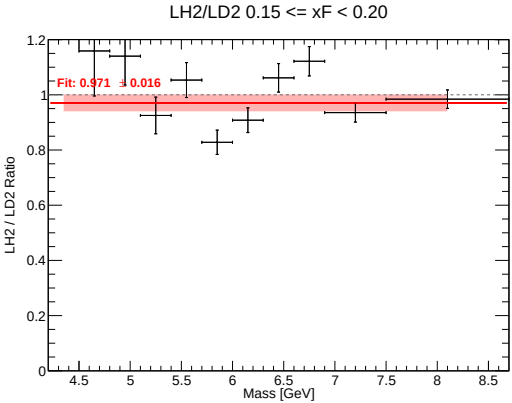
Kinematic Yields Squared (Clean Accepted): x_F Bin 3



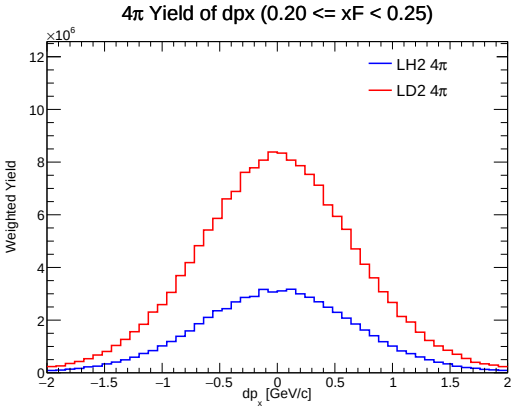


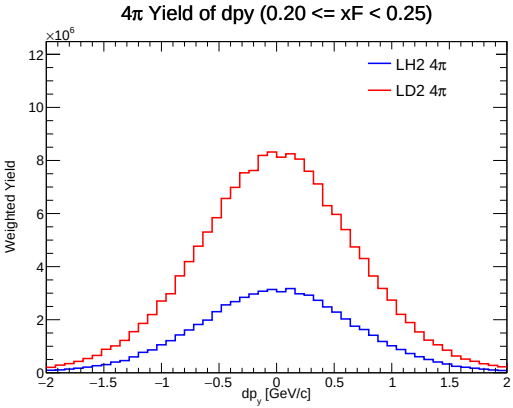
Overlay & Ratio: x_F Bin 3



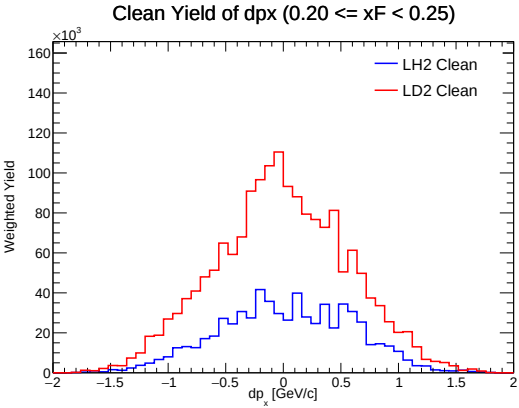


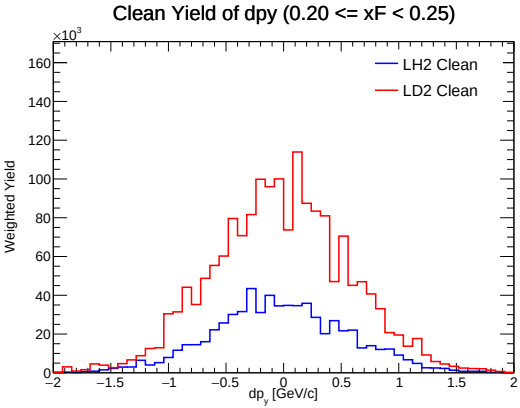
Kinematic Yields (4π Thrown): x_F Bin 4



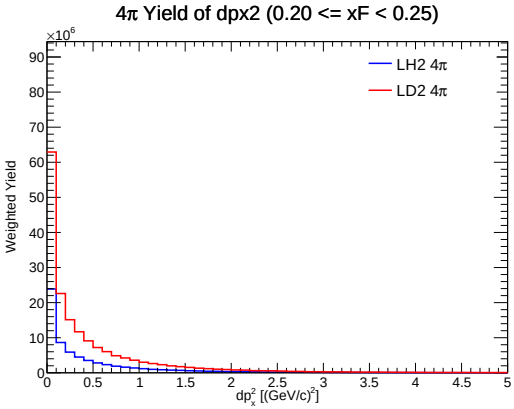


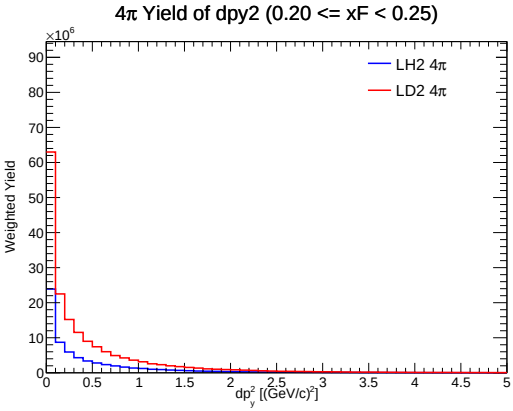
Kinematic Yields (Clean Accepted): x_F Bin 4



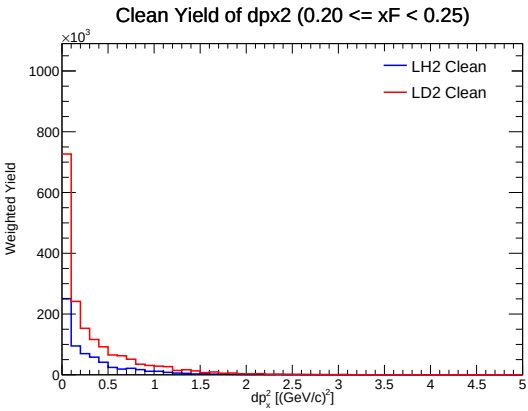


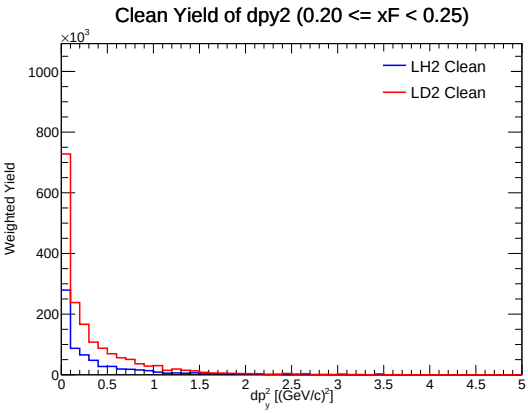
Kinematic Yields Squared (4π Thrown): x_F Bin 4



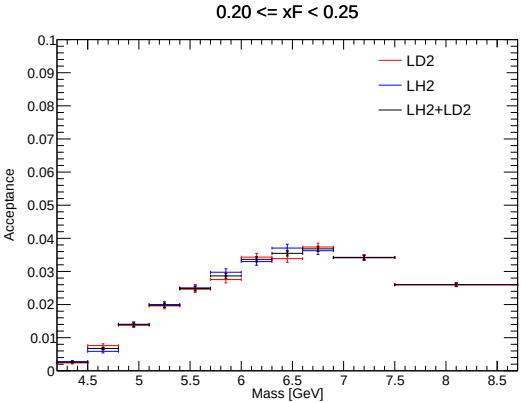


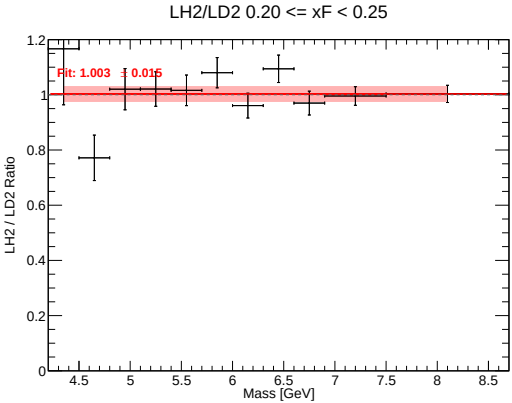
Kinematic Yields Squared (Clean Accepted): x_F Bin 4



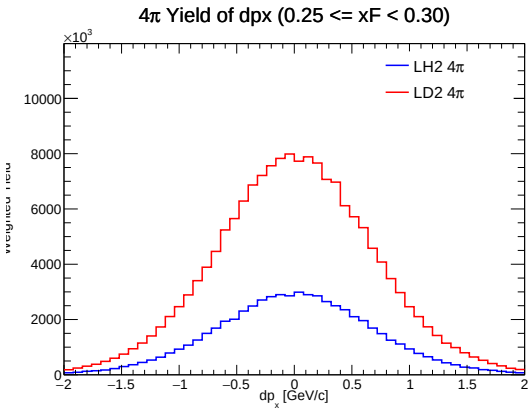


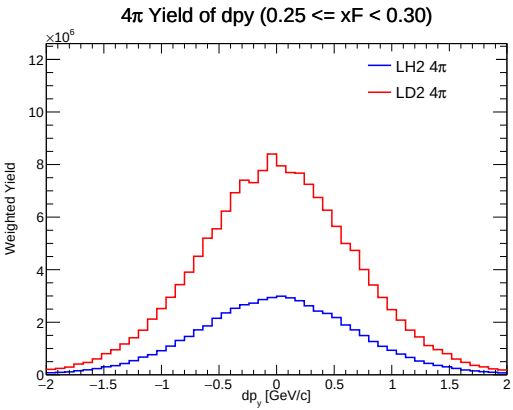
Overlay & Ratio: x_F Bin 4



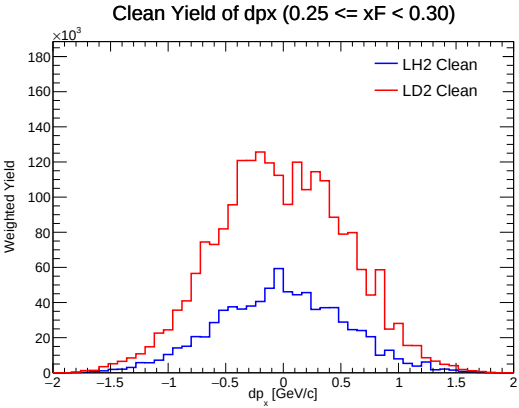


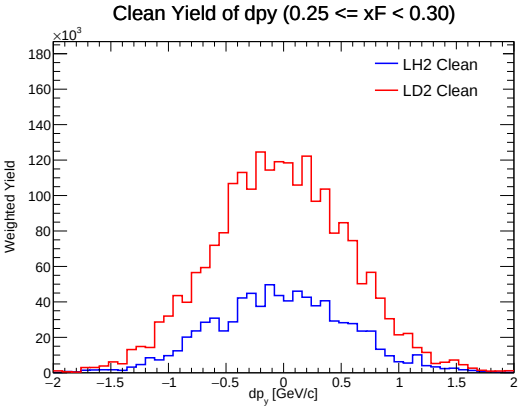
Kinematic Yields (4π Thrown): x_F Bin 5



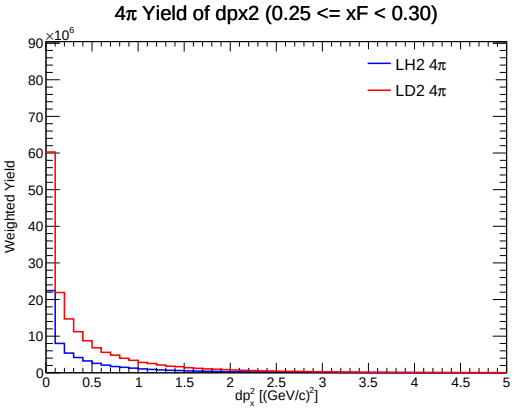


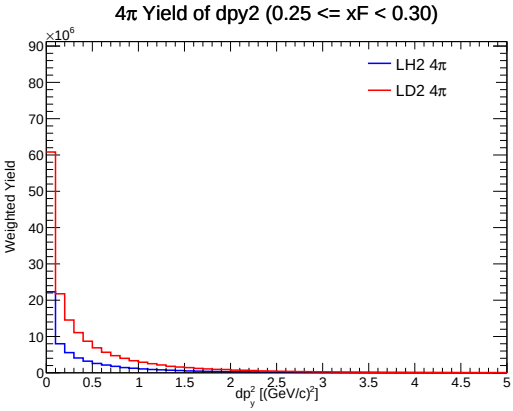
Kinematic Yields (Clean Accepted): x_F Bin 5



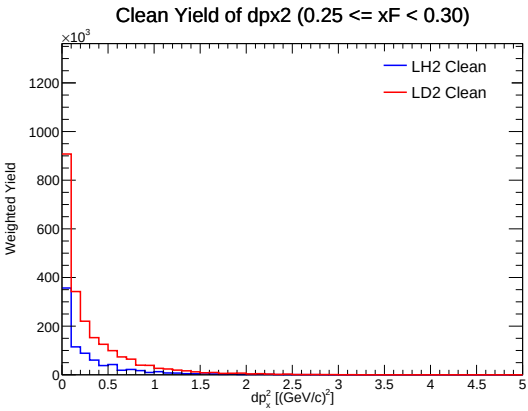


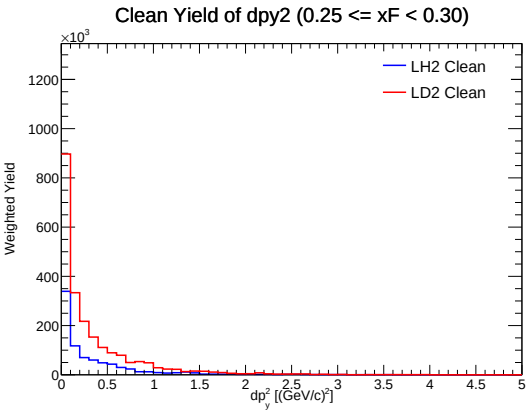
Kinematic Yields Squared (4π Thrown): x_F Bin 5



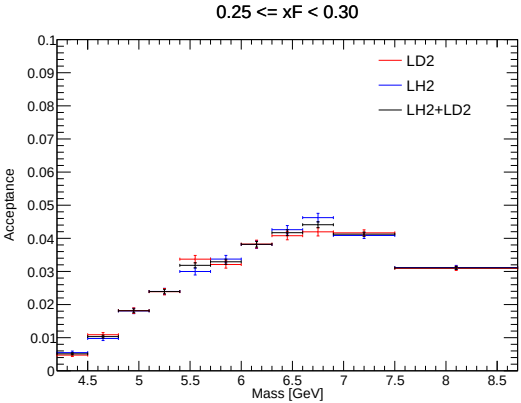


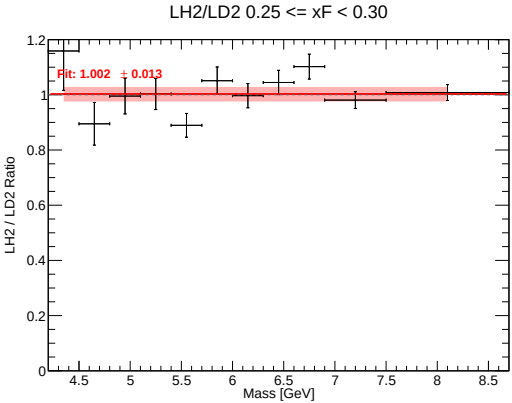
Kinematic Yields Squared (Clean Accepted): x_F Bin 5



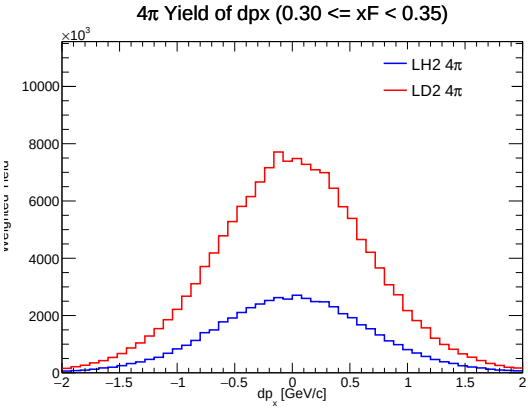


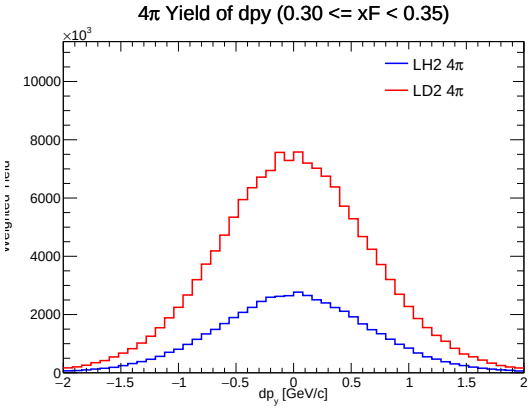
Overlay & Ratio: x_F Bin 5



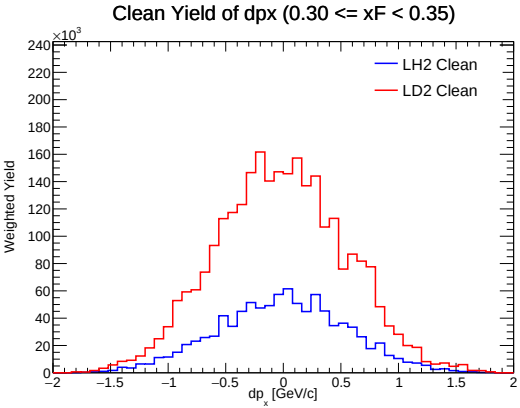


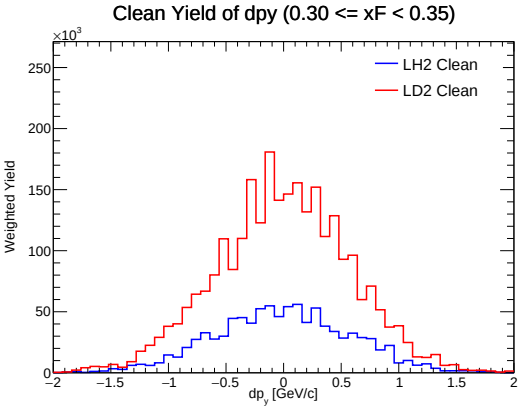
Kinematic Yields (4π Thrown): x_F Bin 6



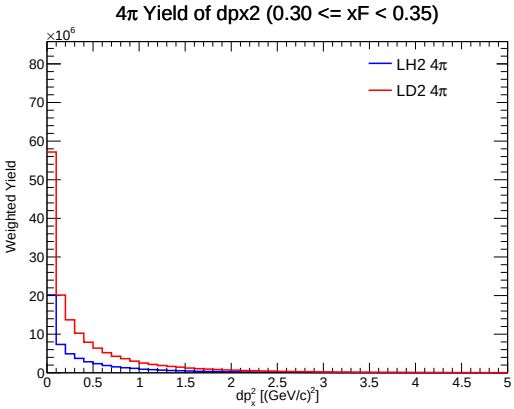


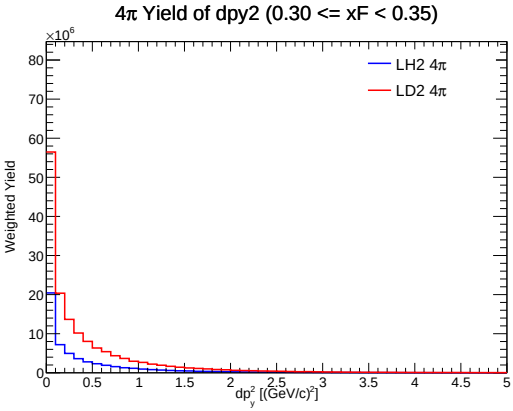
Kinematic Yields (Clean Accepted): x_F Bin 6



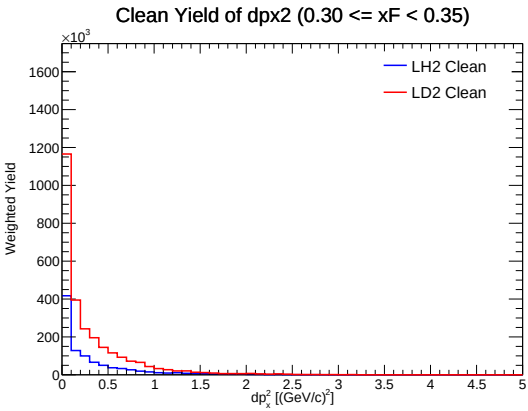


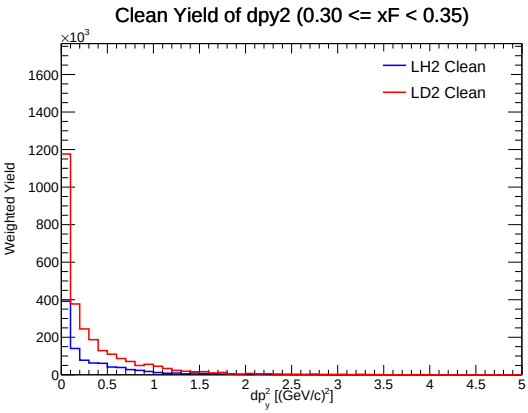
Kinematic Yields Squared (4π Thrown): x_F Bin 6



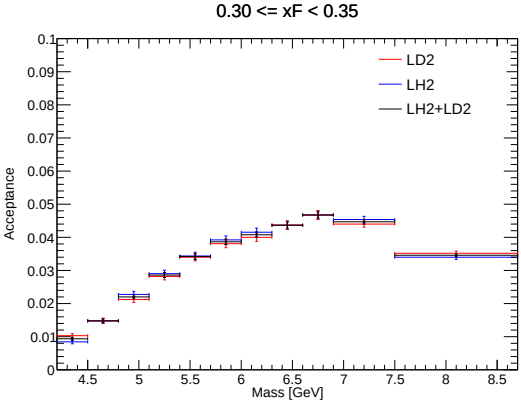


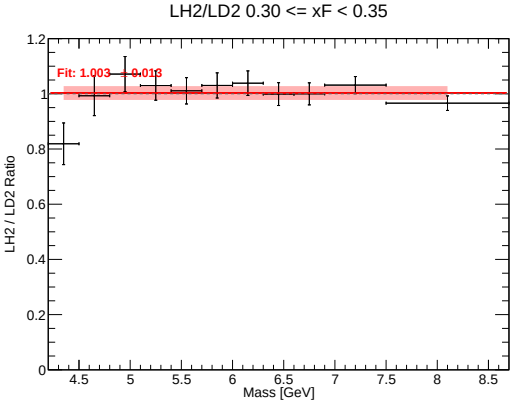
Kinematic Yields Squared (Clean Accepted): x_F Bin 6



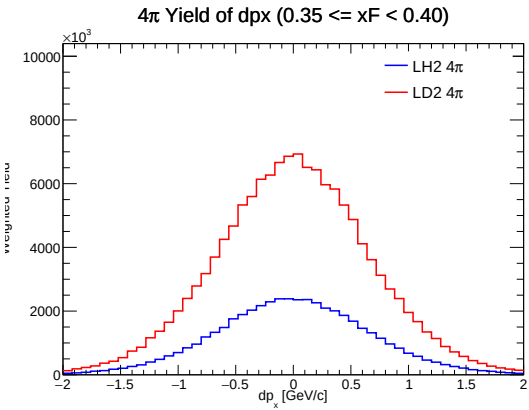


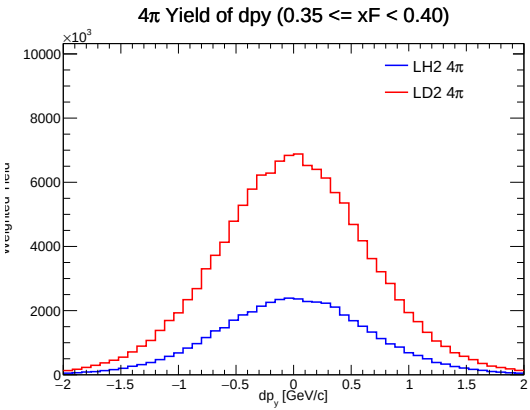
Overlay & Ratio: x_F Bin 6



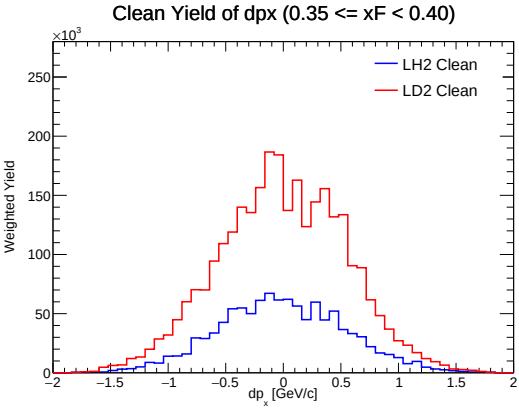


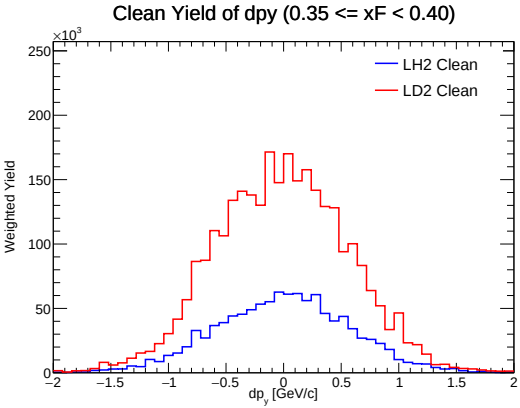
Kinematic Yields (4π Thrown): x_F Bin 7



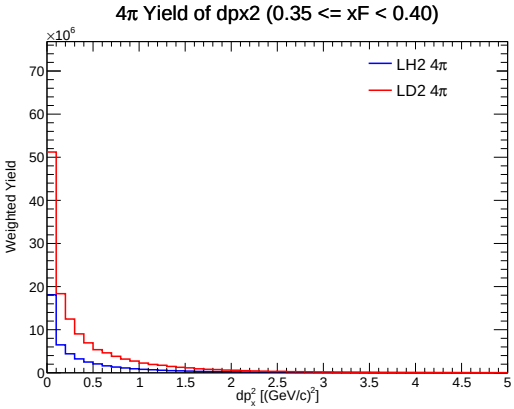


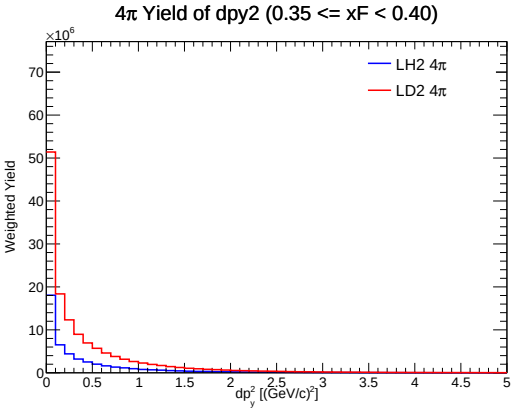
Kinematic Yields (Clean Accepted): x_F Bin 7



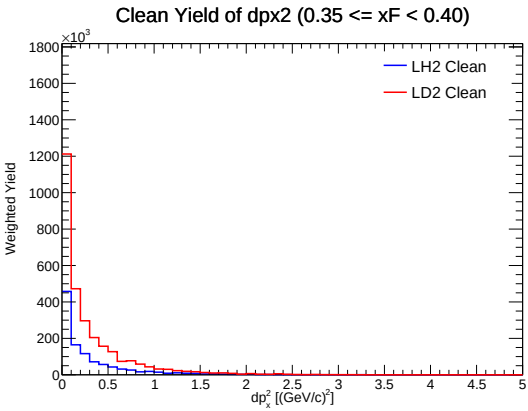


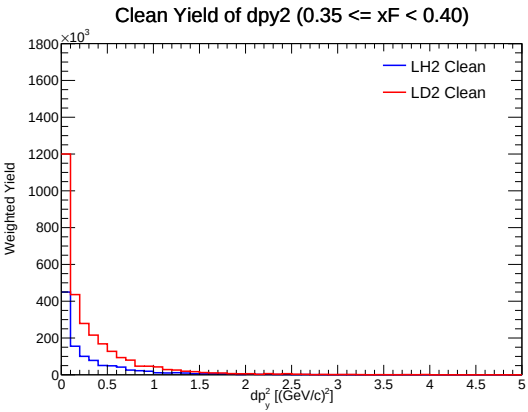
Kinematic Yields Squared (4π Thrown): x_F Bin 7



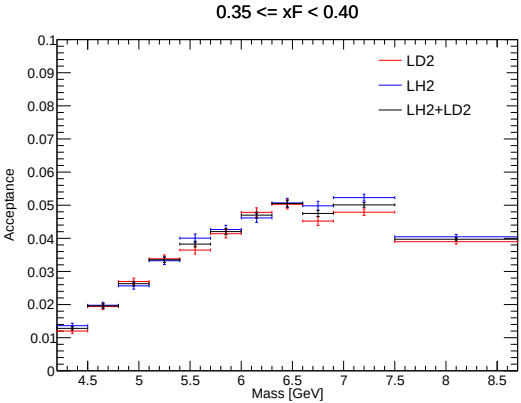


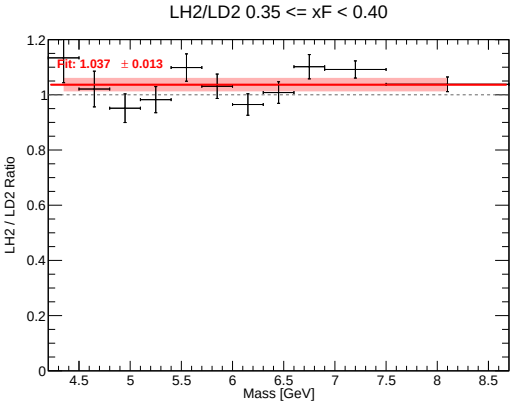
Kinematic Yields Squared (Clean Accepted): x_F Bin 7



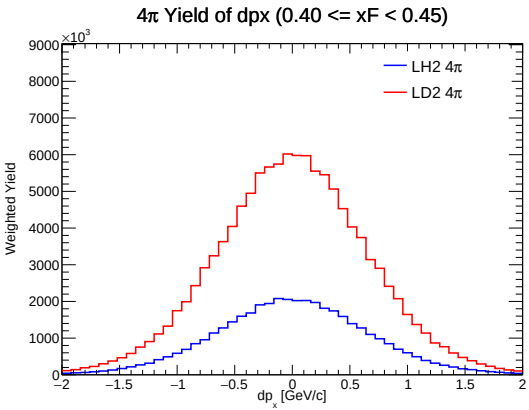


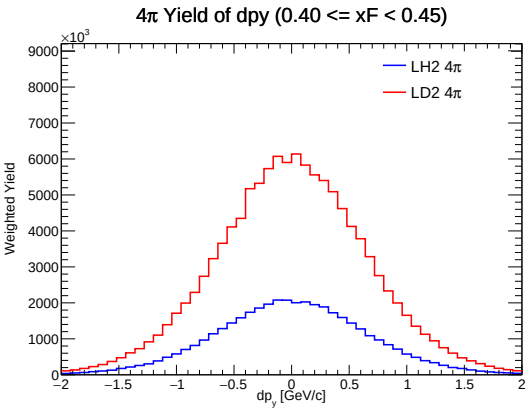
Overlay & Ratio: x_F Bin 7



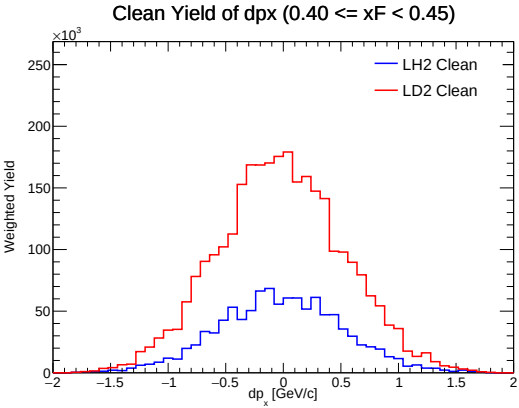


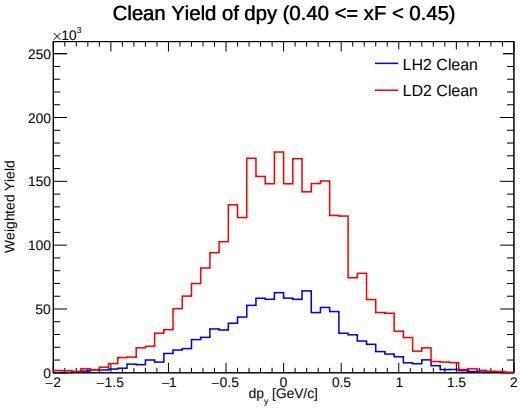
Kinematic Yields (4π Thrown): x_F Bin 8



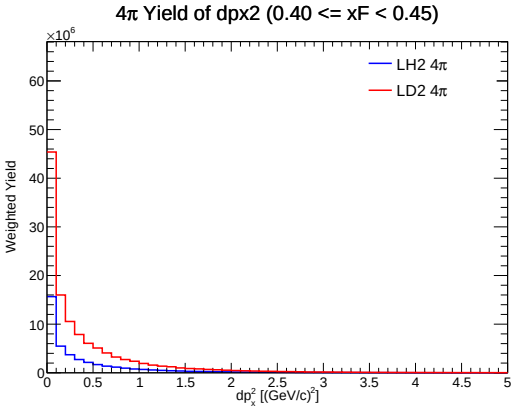


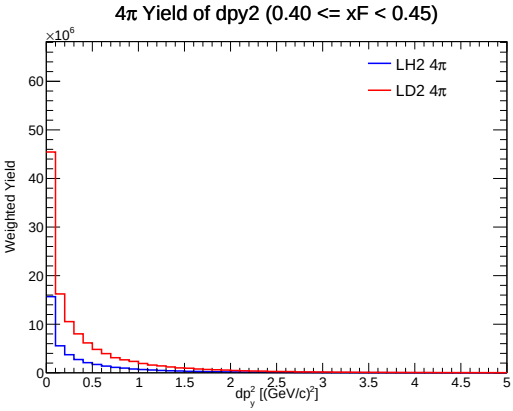
Kinematic Yields (Clean Accepted): x_F Bin 8



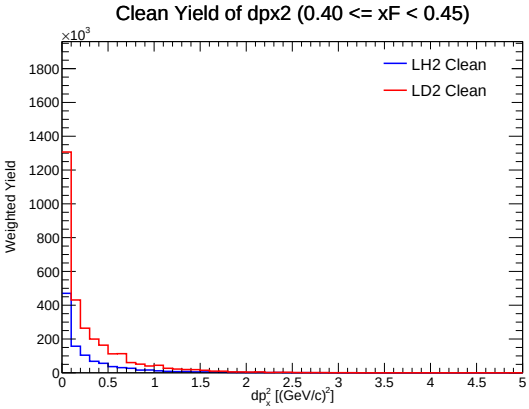


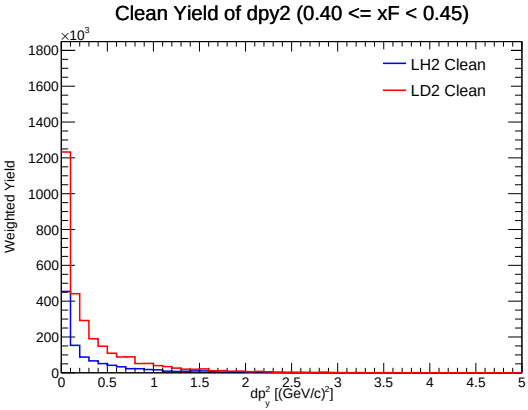
Kinematic Yields Squared (4π Thrown): x_F Bin 8



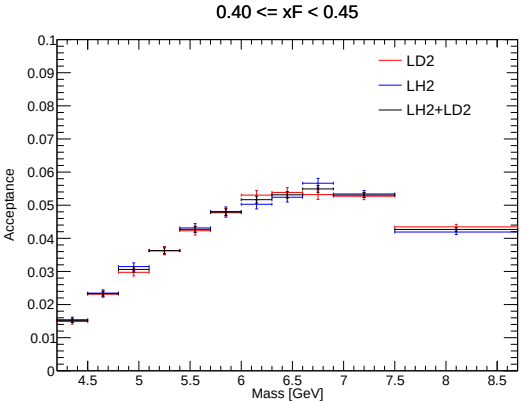


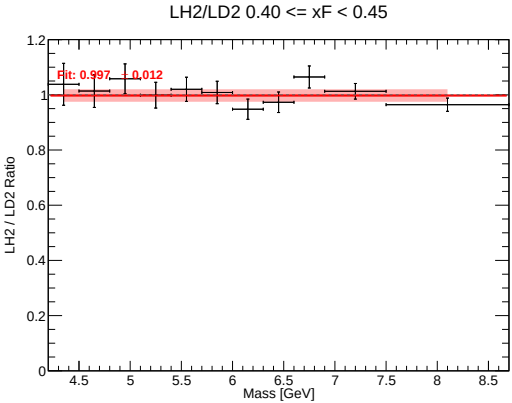
Kinematic Yields Squared (Clean Accepted): x_F Bin 8



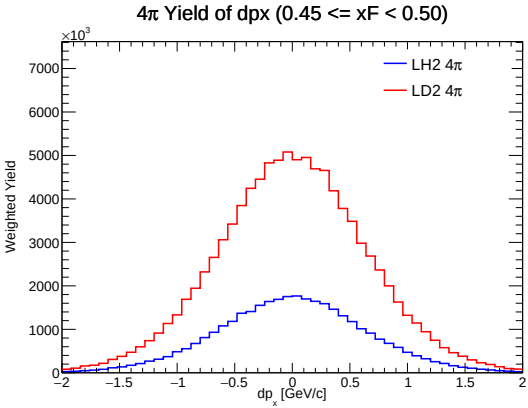


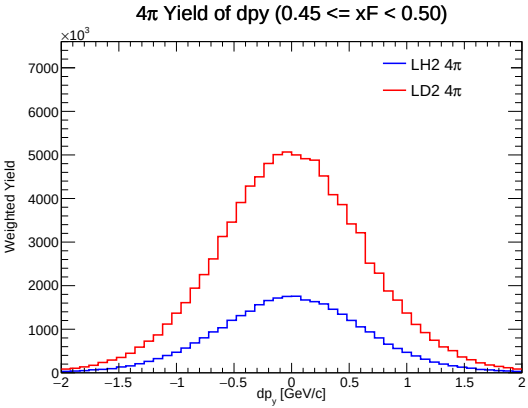
Overlay & Ratio: x_F Bin 8



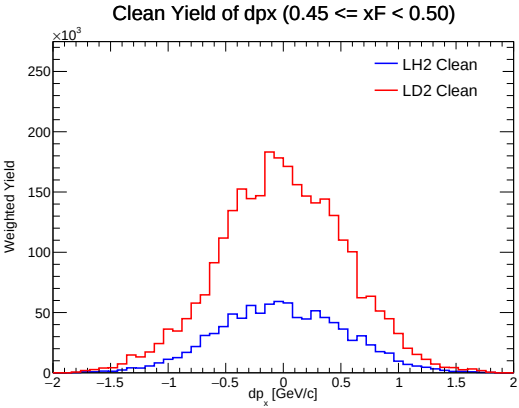


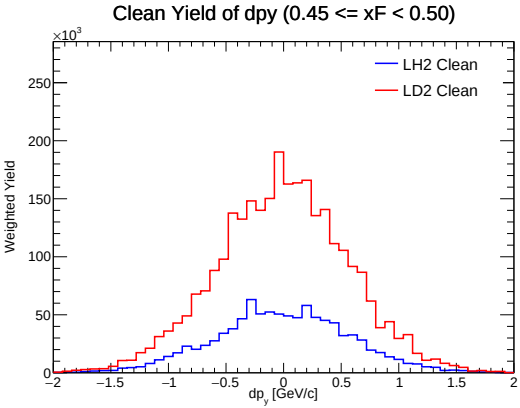
Kinematic Yields (4π Thrown): x_F Bin 9



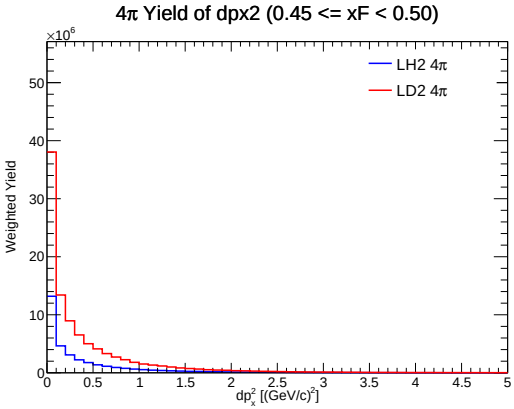


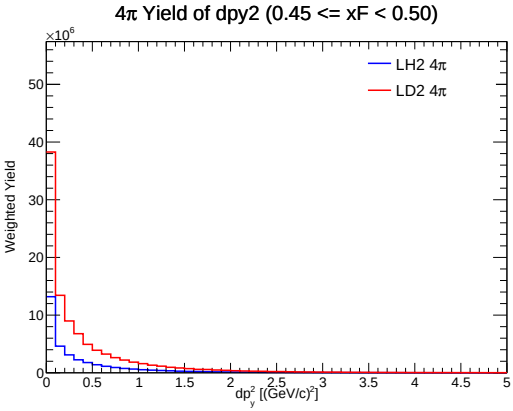
Kinematic Yields (Clean Accepted): x_F Bin 9



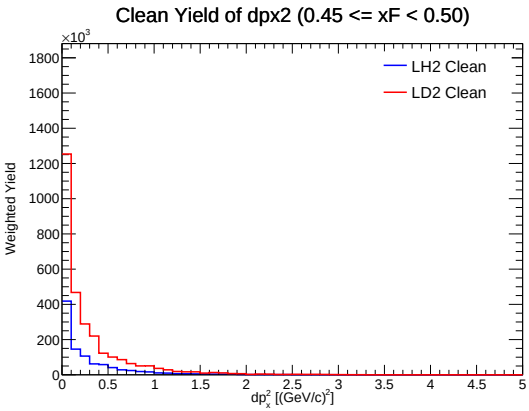


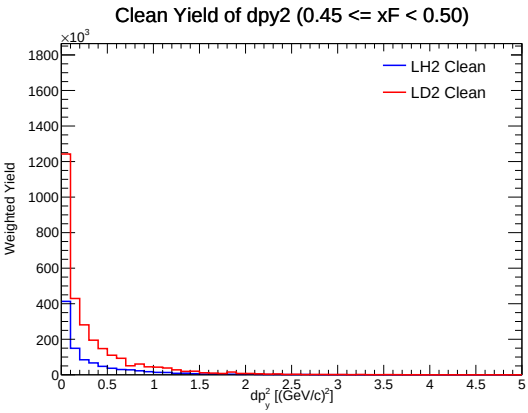
Kinematic Yields Squared (4π Thrown): x_F Bin 9



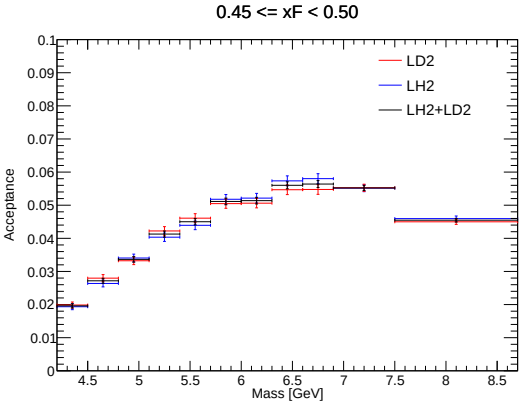


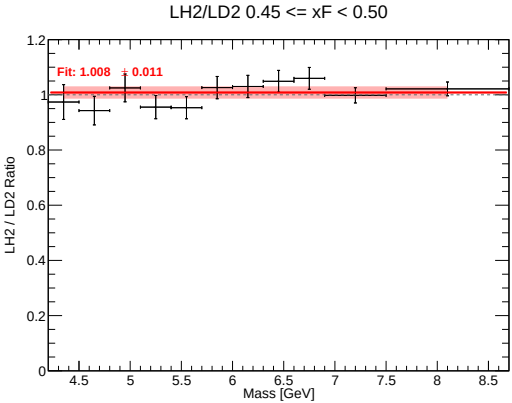
Kinematic Yields Squared (Clean Accepted): x_F Bin 9



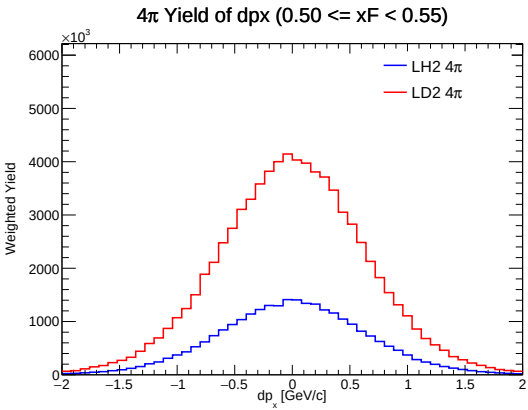


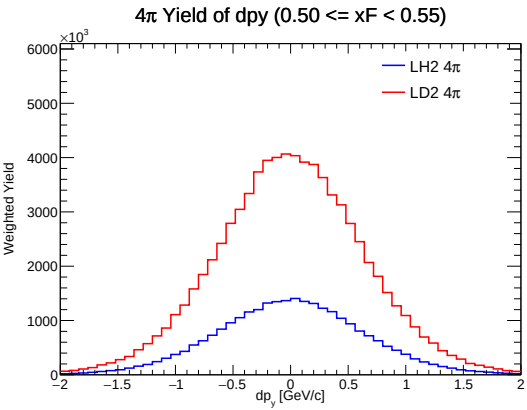
Overlay & Ratio: x_F Bin 9



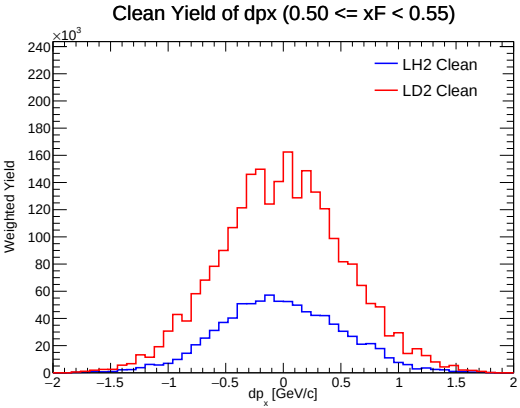


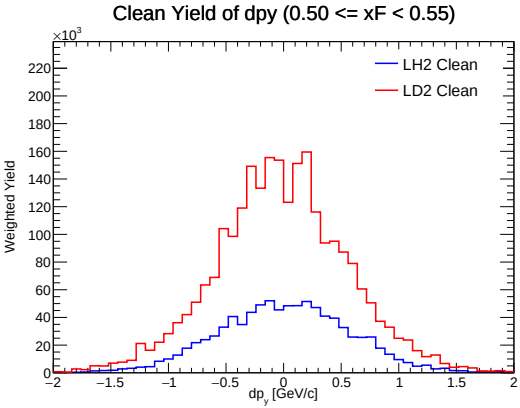
Kinematic Yields (4π Thrown): x_F Bin 10



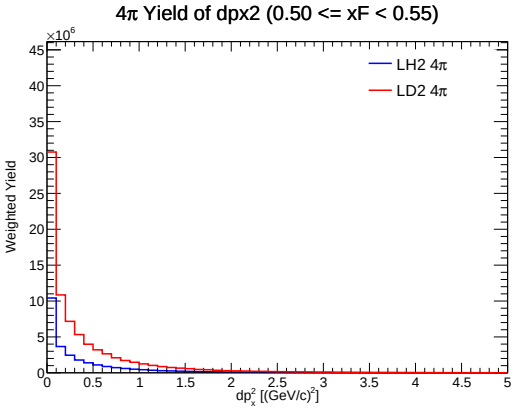


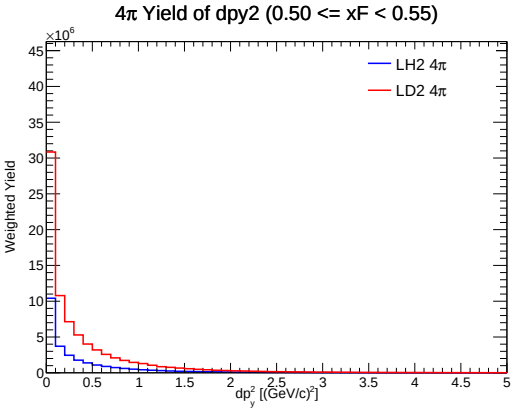
Kinematic Yields (Clean Accepted): x_F Bin 10



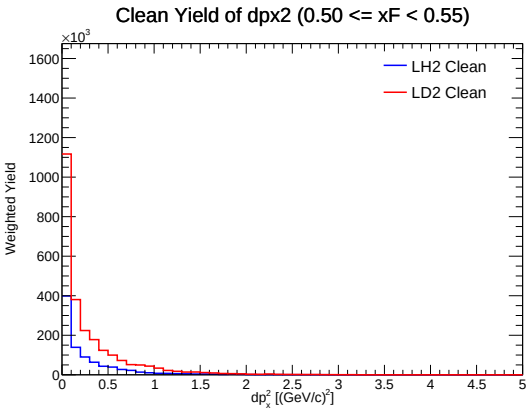


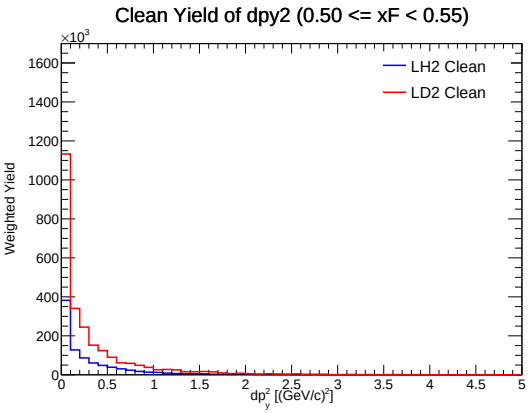
Kinematic Yields Squared (4π Thrown): x_F Bin 10



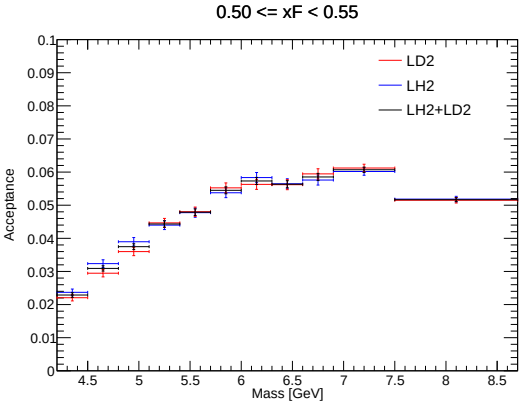


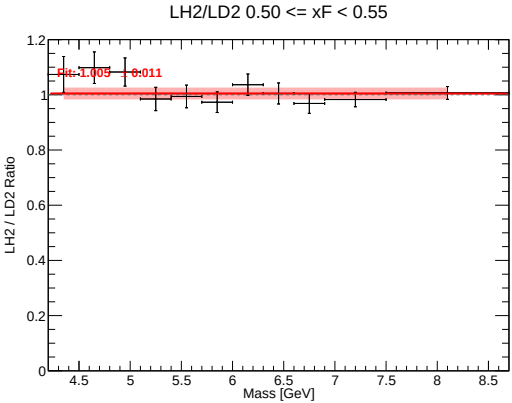
Kinematic Yields Squared (Clean Accepted): x_F Bin 10



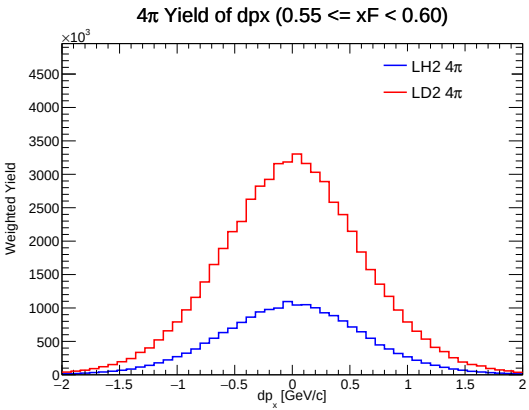


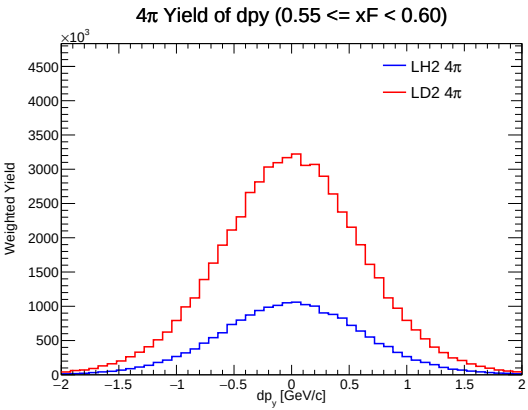
Overlay & Ratio: x_F Bin 10



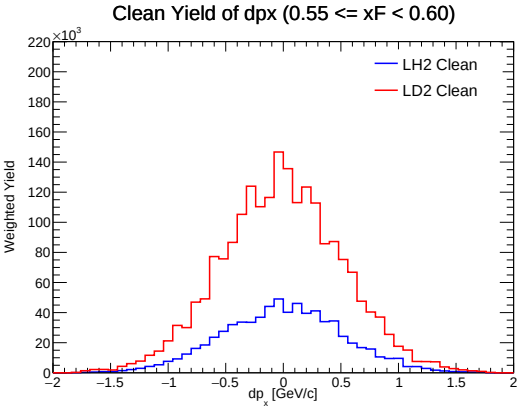


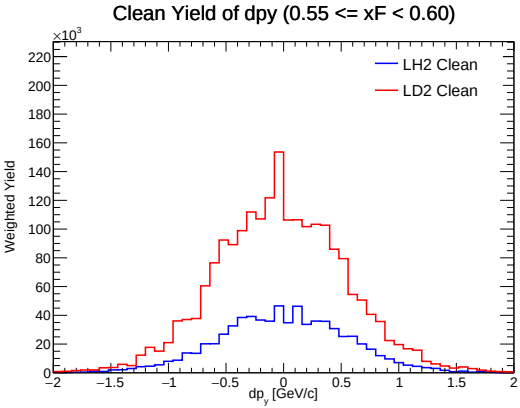
Kinematic Yields (4π Thrown): x_F Bin 11



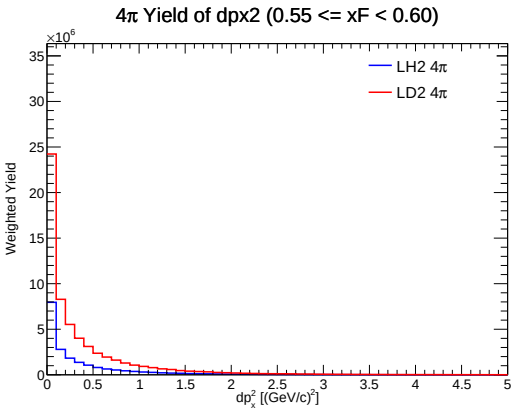


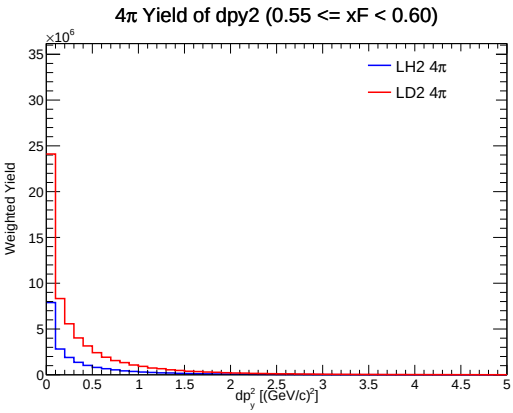
Kinematic Yields (Clean Accepted): x_F Bin 11



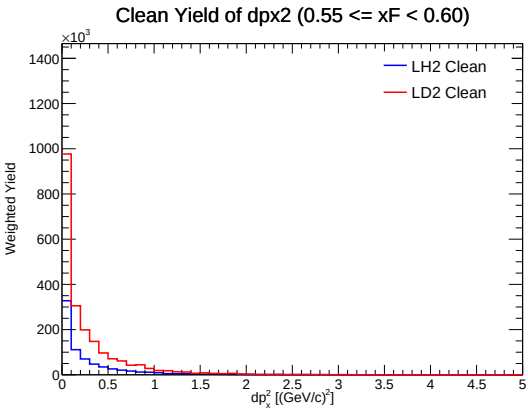


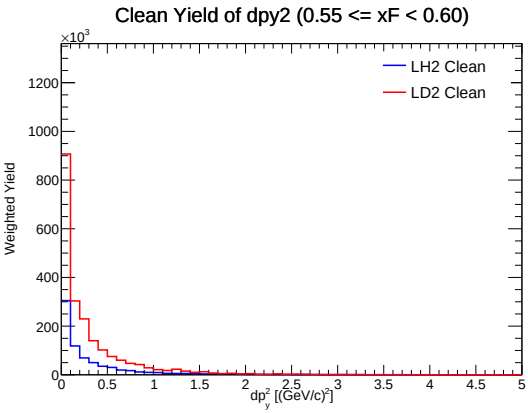
Kinematic Yields Squared (4π Thrown): x_F Bin 11



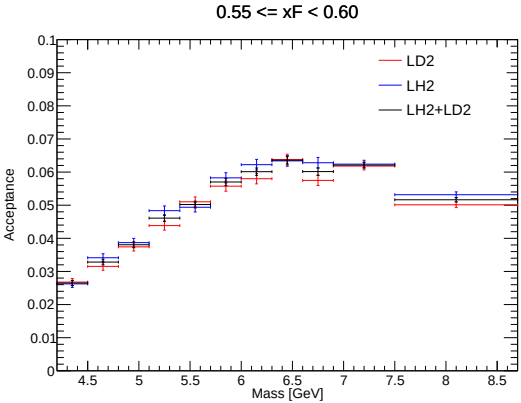


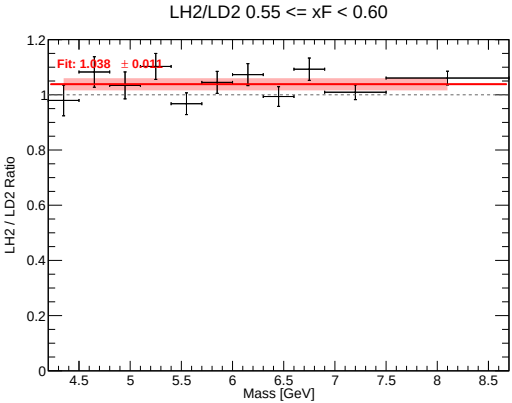
Kinematic Yields Squared (Clean Accepted): x_F Bin 11



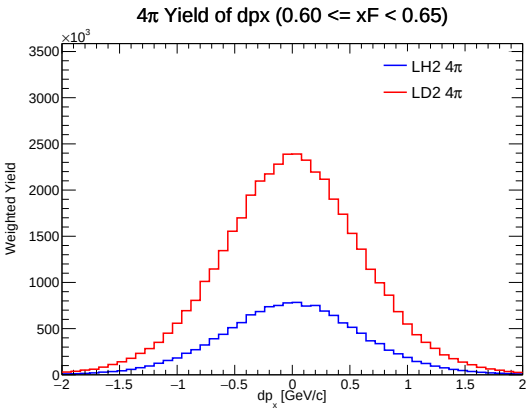


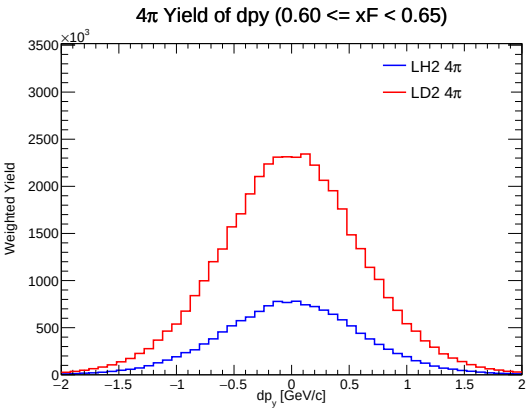
Overlay & Ratio: x_F Bin 11



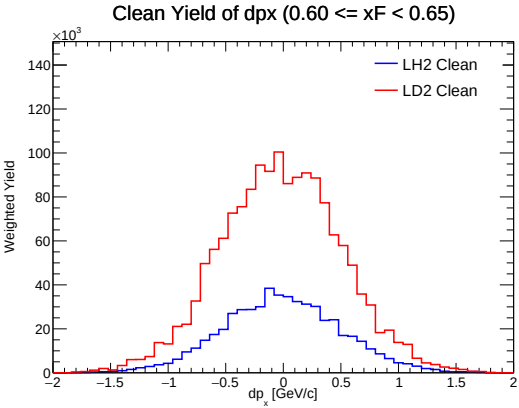


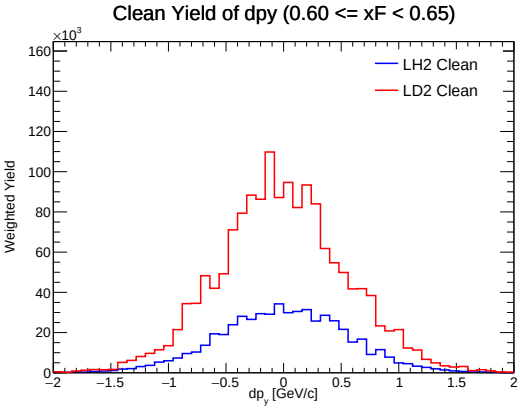
Kinematic Yields (4π Thrown): x_F Bin 12



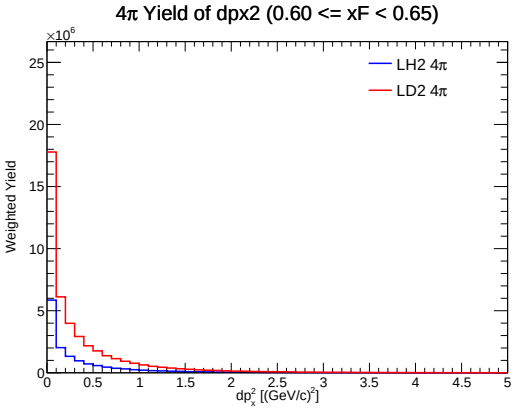


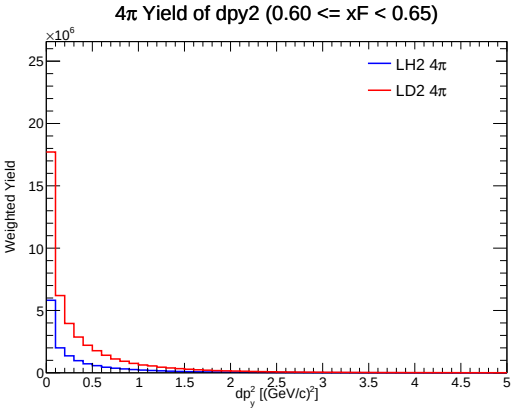
Kinematic Yields (Clean Accepted): x_F Bin 12



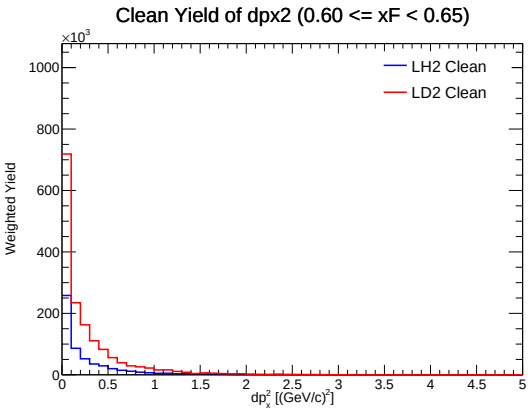


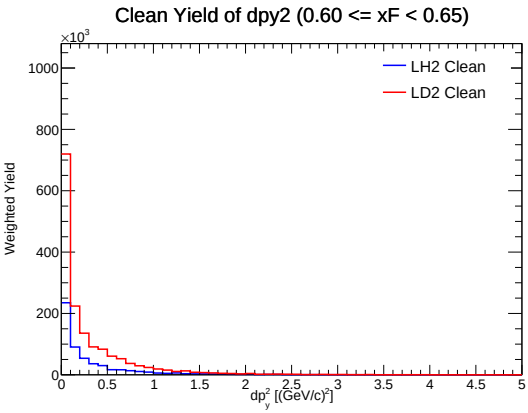
Kinematic Yields Squared (4π Thrown): x_F Bin 12



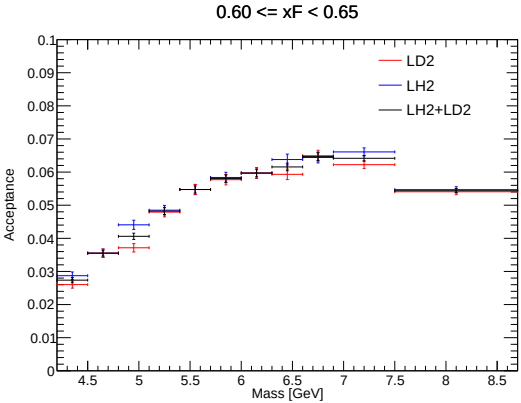


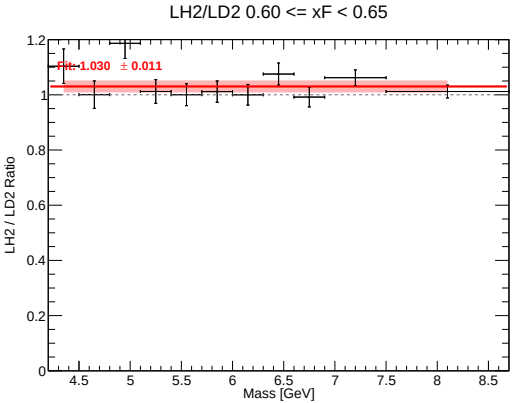
Kinematic Yields Squared (Clean Accepted): x_F Bin 12



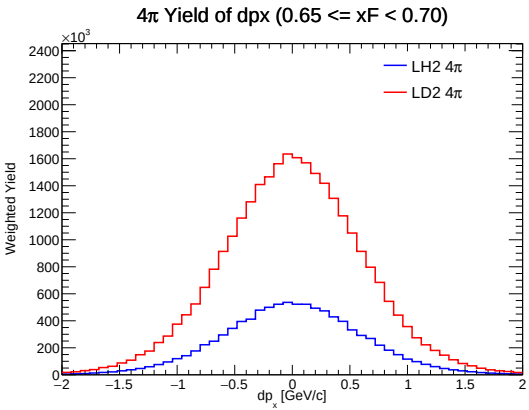


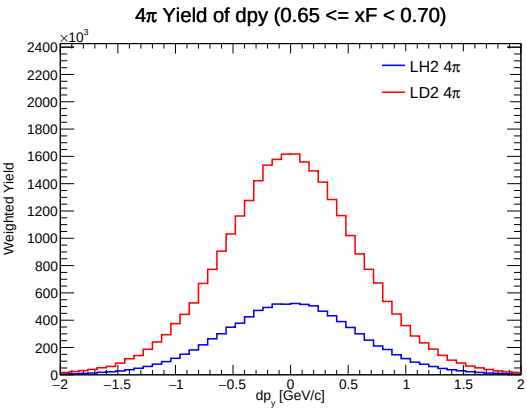
Overlay & Ratio: x_F Bin 12



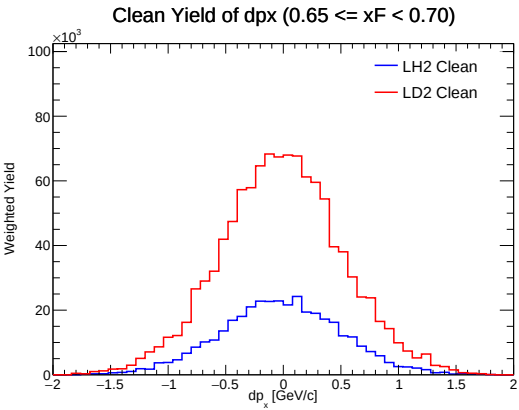


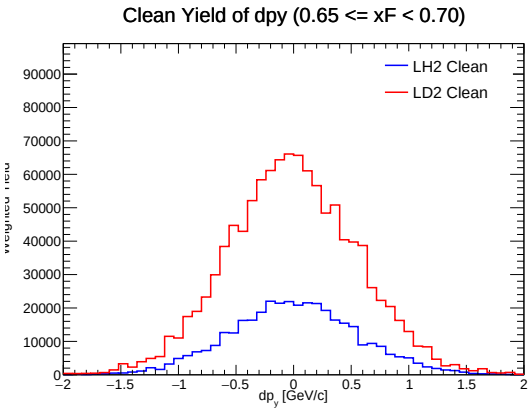
Kinematic Yields (4π Thrown): x_F Bin 13



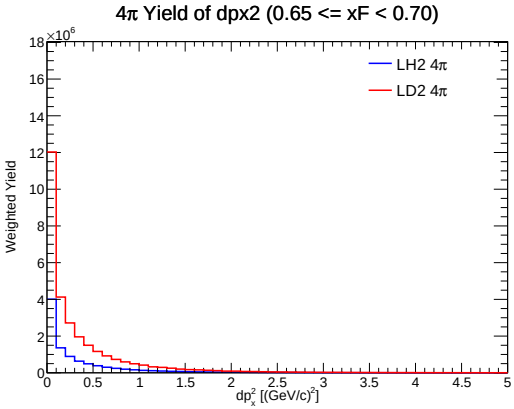


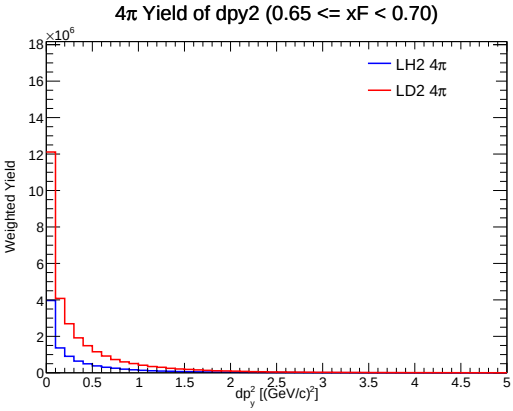
Kinematic Yields (Clean Accepted): x_F Bin 13



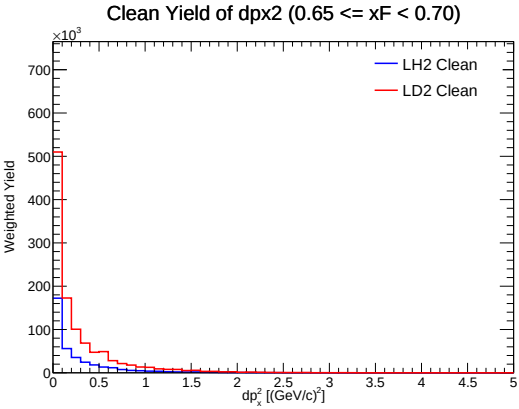


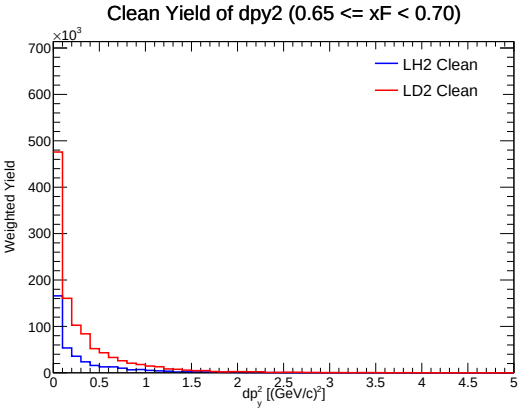
Kinematic Yields Squared (4π Thrown): x_F Bin 13



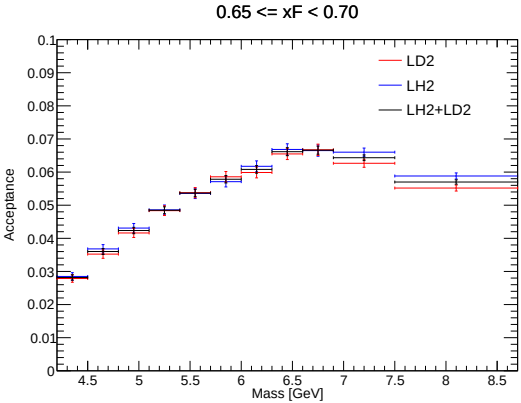


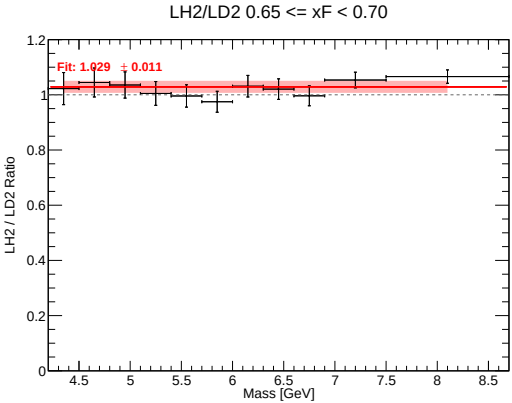
Kinematic Yields Squared (Clean Accepted): x_F Bin 13



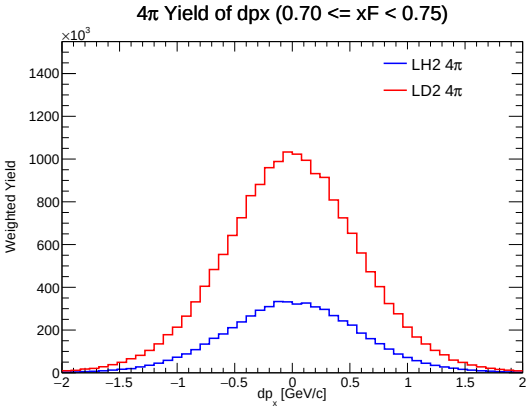


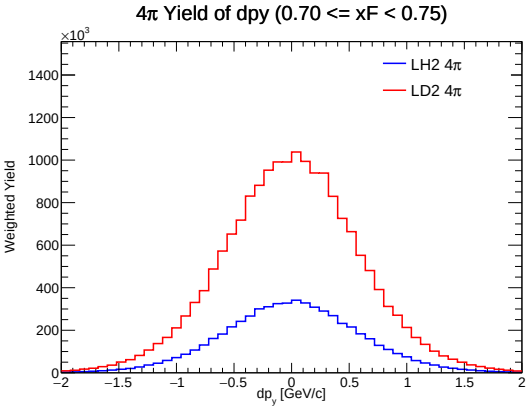
Overlay & Ratio: x_F Bin 13



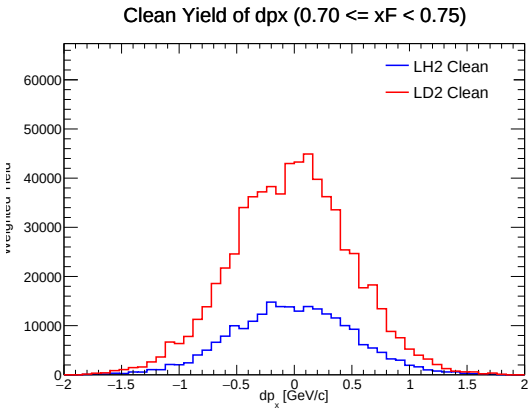


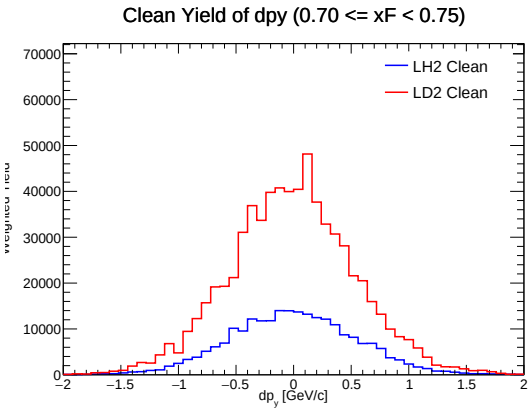
Kinematic Yields (4π Thrown): x_F Bin 14



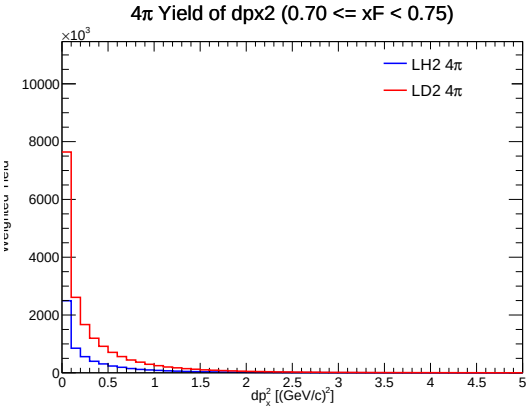


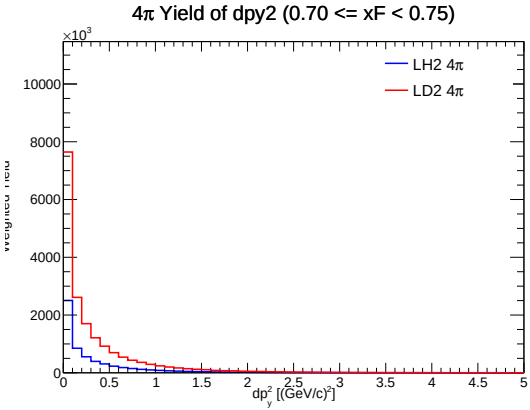
Kinematic Yields (Clean Accepted): x_F Bin 14



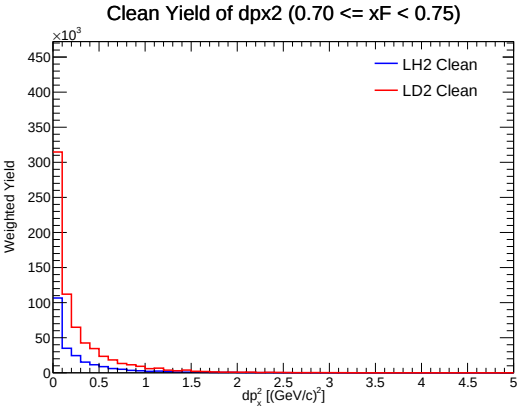


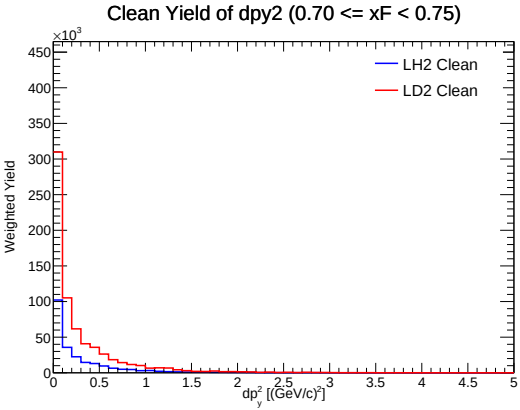
Kinematic Yields Squared (4π Thrown): x_F Bin 14



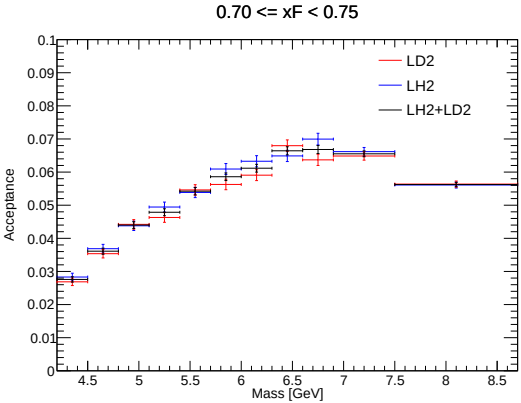


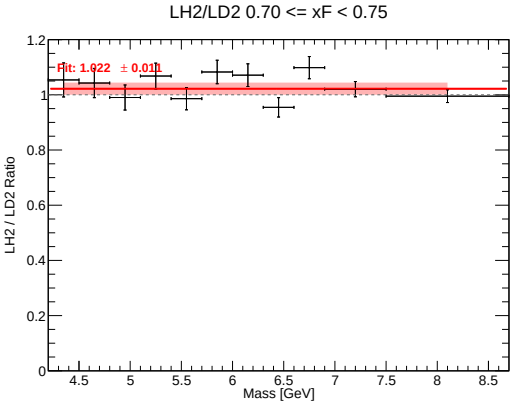
Kinematic Yields Squared (Clean Accepted): x_F Bin 14



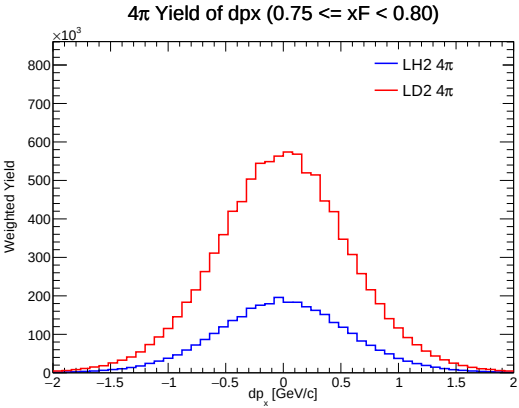


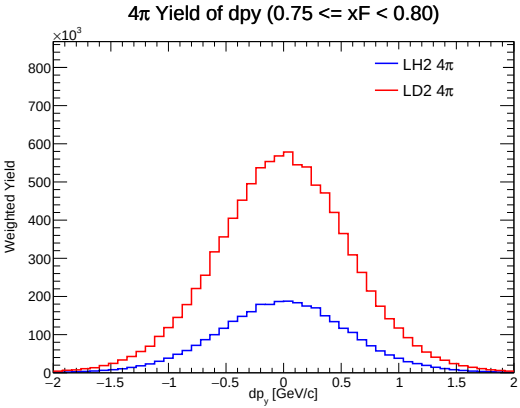
Overlay & Ratio: x_F Bin 14



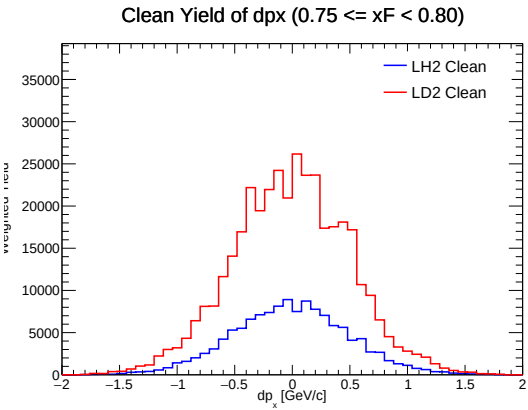


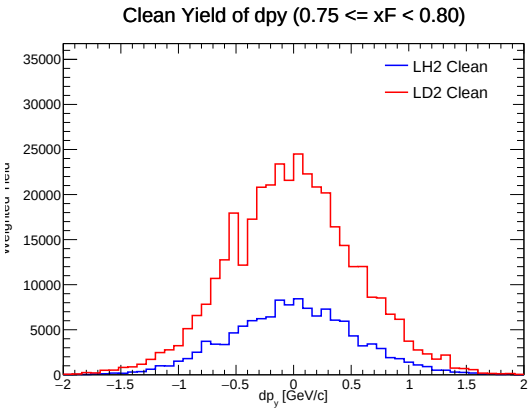
Kinematic Yields (4π Thrown): x_F Bin 15



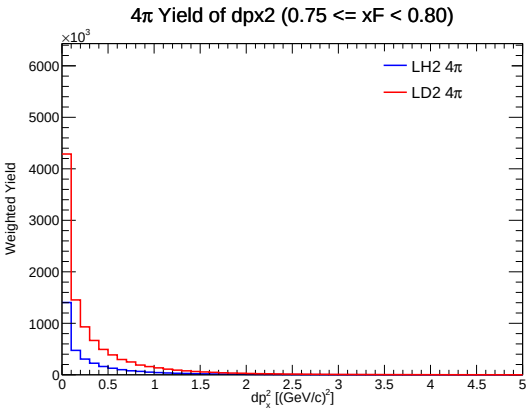


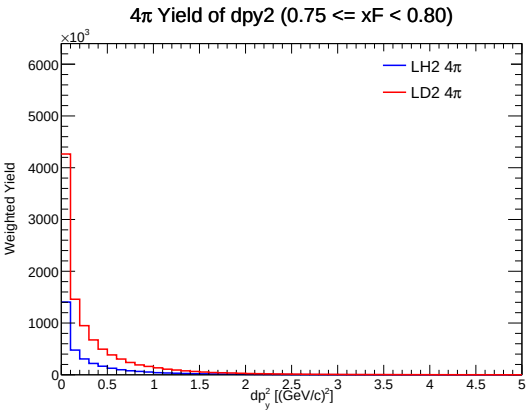
Kinematic Yields (Clean Accepted): x_F Bin 15



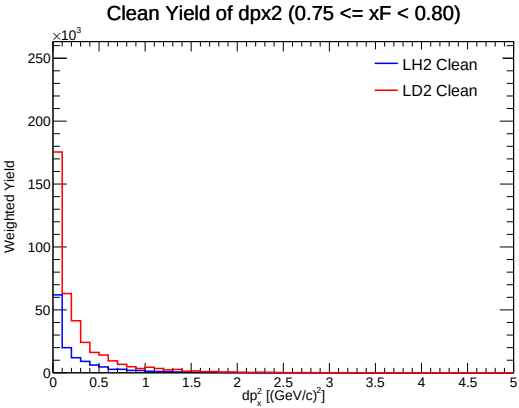


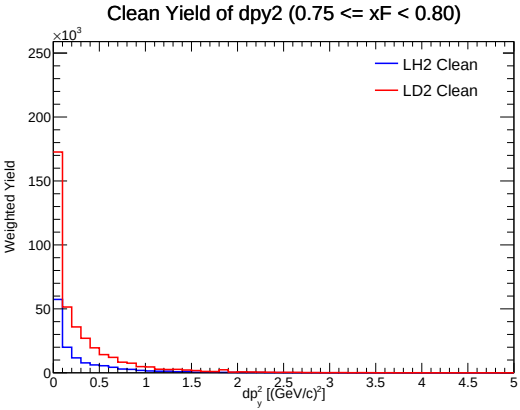
Kinematic Yields Squared (4π Thrown): x_F Bin 15



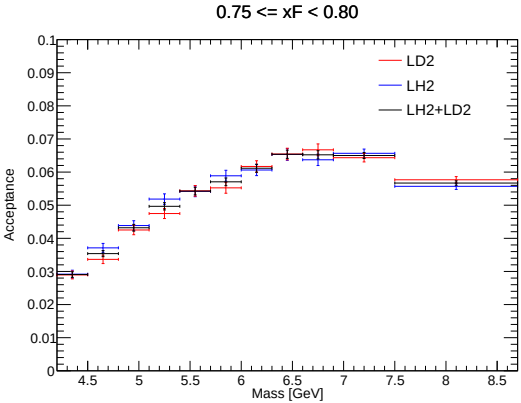


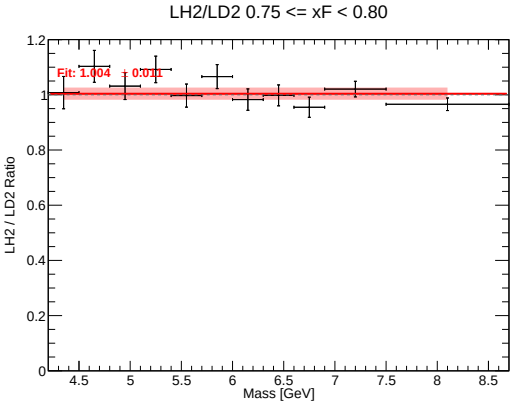
Kinematic Yields Squared (Clean Accepted): x_F Bin 15



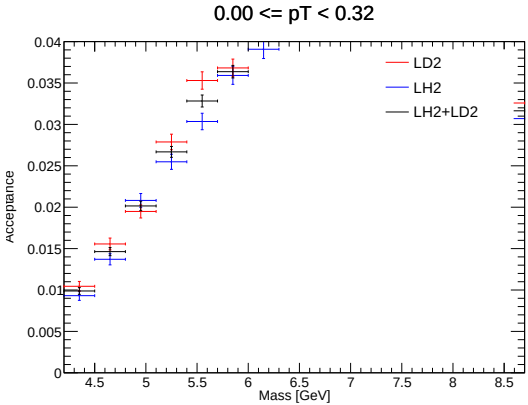


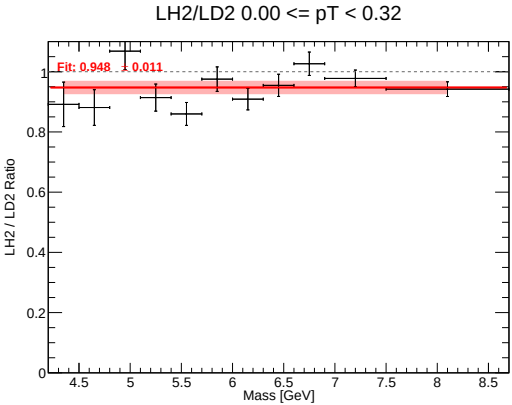
Overlay & Ratio: x_F Bin 15



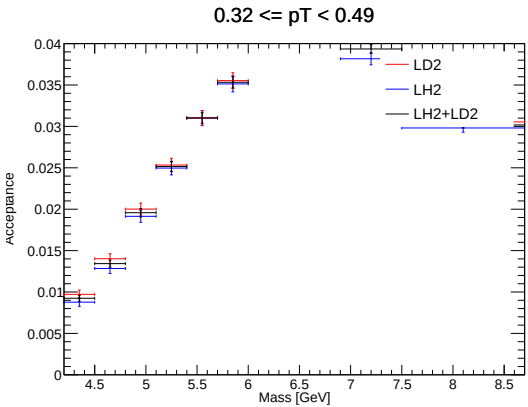


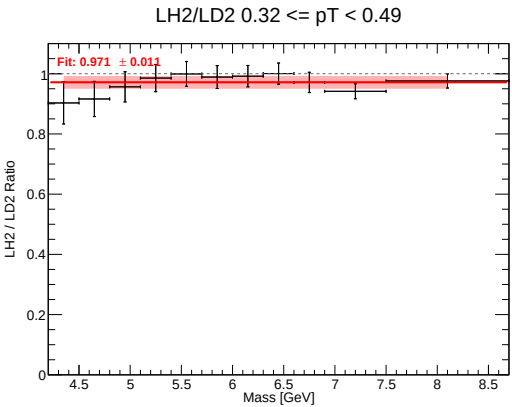
Overlay & Ratio: p_T Bin 0



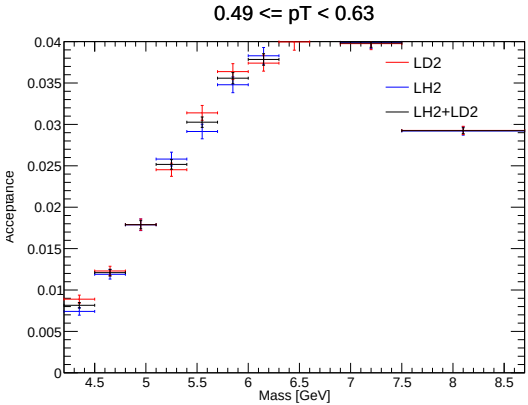


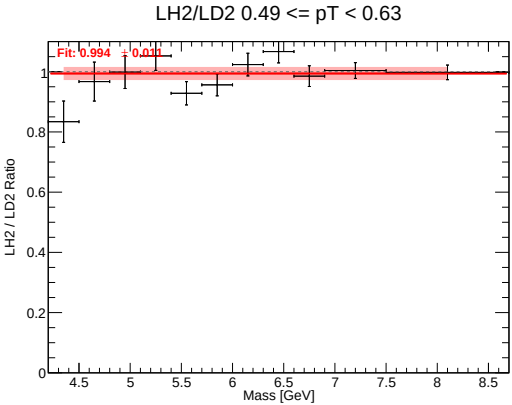
Overlay & Ratio: p_T Bin 1



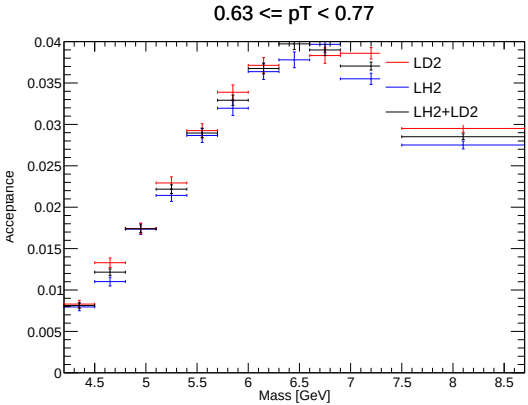


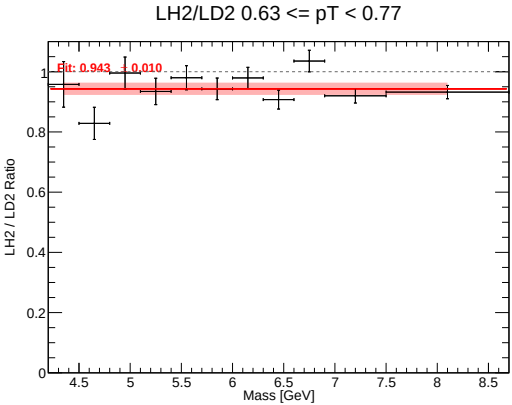
Overlay & Ratio: p_T Bin 2



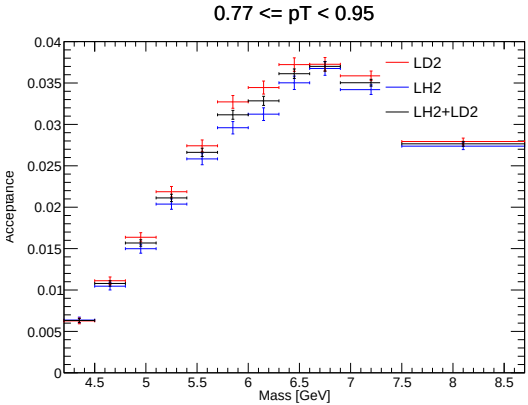


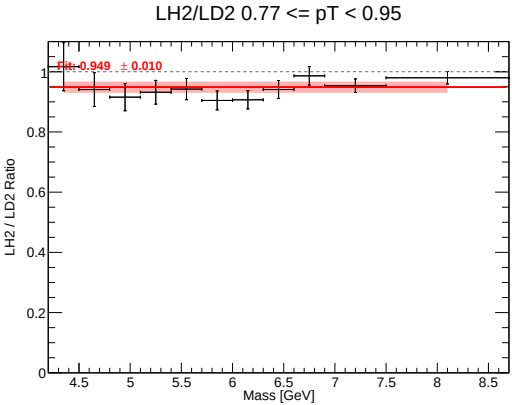
Overlay & Ratio: p_T Bin 3



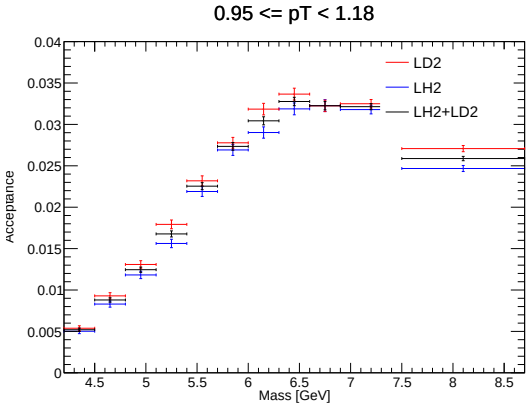


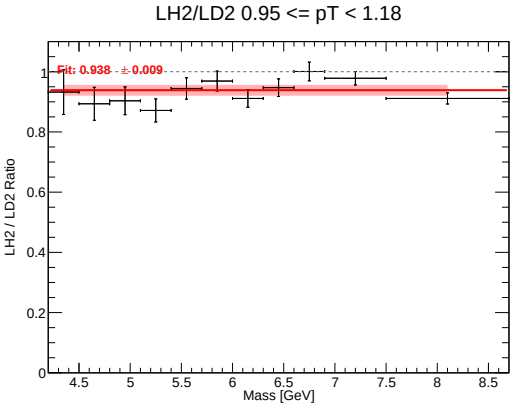
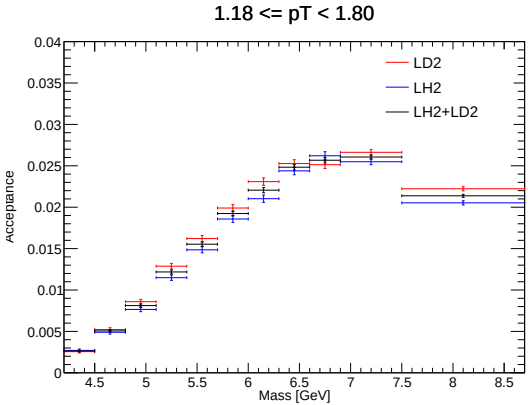
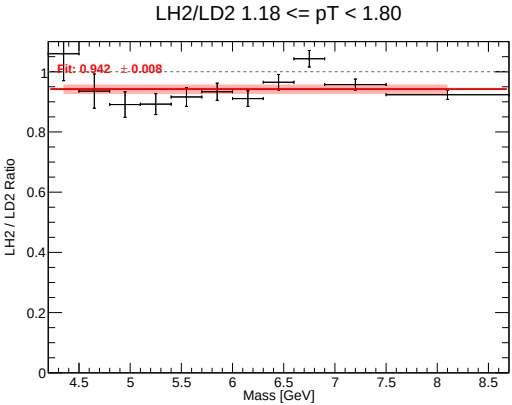
Overlay & Ratio: p_T Bin 4





Overlay & Ratio: p_T Bin 5





Overlay & Ratio: p_T Bin 6